

Spectrum Virtualize 3-Site Replication

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 **Cloud**

Storage



International Technical Support Organization

Spectrum Virtualize 3-Site Replication

March 2020

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First Edition (March 2020)

This edition applies to Spectrum Virtualize V8.3.1.1 and the other software and hardware products described in this book. Contact your IBM representative if you are unsure as to its applicability to your environment.

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Preface

In this IBM® Redbooks® publication we describe IBM Spectrum® Virtualize 3-Site Replication and its implementation.

Note: At the time of writing, 3-Site Replication Orchestration software is available using the Storage Customer Opportunity REquest (SCORE)/Request for Price Quotations (RPQ).

A SCORE/RPQ should be submitted to IBM requesting approval via your IBM Representative. The SCORE request should include configuration details requested in the 3-Site Replication questionnaire.

[Spectrum Virtualize 3-Site Replication Questionnaire](#)

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3-Site Replication concepts

This chapter gives a general overview of the 3-Site Replication feature and its architecture. Existing Spectrum Virtualize features are used to implement this feature. This chapter covers the following topics:

- ▶ How 3-Site Replication works
- ▶ Metro Mirror and Remote Copy Consistency Groups
- ▶ 3-Site Replication: mirroring and replication
- ▶ 3-Site Replication: Consistency Groups
- ▶ Data consistency
- ▶ 3-Site user

Note: At the time of writing, 3-Site Replication Orchestration software is available using the Storage Customer Opportunity Request (SCORE)/Request for Price Quotations (RPQ).

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[Spectrum Virtualize 3-Site Replication Questionnaire](#)

1.1 How 3-Site Replication works

This section describes the *IBM Spectrum Virtualize 3-Site Replication* feature. Clients use the ability to maintain two independent copies of a single data volume across two sites, using the Spectrum Virtualize Metro Mirror feature. The data is replicated synchronously from a Spectrum Virtualize cluster at a first site to another Spectrum Virtualize cluster in a second site. These two sites are independent from each other and build two separate failure domains. If one site fails then the other site still has the same data as the failing site.

Starting with Spectrum Virtualize V8.3.1.1, two site replication can be extended to a third site. Using a three site setup means data can be replicated asynchronously to an independent system in a third site. The third site can be far away from the first or second site. Therefore the replication uses asynchronous replication techniques. This three site setup is called *IBM Spectrum Virtualize 3-Site Replication*.

3-Site Replication is managed, operated, and maintained by a dedicated software which is called *Orchestrator*. Orchestrator runs on a separate Linux system and is independent from the Spectrum Virtualize software.

Some of the 3-Site Replication properties are:

- ▶ Three independent Spectrum Virtualize clusters. An outage of one cluster does not affect the accessibility of another cluster.
- ▶ Two sites replicate volume data synchronously.
- ▶ Volume data is replicated asynchronously to the third site.
- ▶ Dedicated Orchestrator software assures operation of 3-Site Replication independently from the status and accessibility of any 3-Site Replication cluster.

Figure 1-1 shows an overview of a 3-Site Replication setup. The sites, the active and the standby link to the third site, and the Orchestrator concepts will be discussed throughout this chapter. Replication to the third site is using the active link. The active and the standby link can be switched as discussed in Chapter 5, “3-Site Replication Monitoring and Maintenance” on page 45. 3-Site Replication is based on Spectrum Virtualize remote copy Consistency Groups.

In Figure 1-1 we show a simplified overview to demonstrate the three site concept, and show only one volume of a Consistency Group.

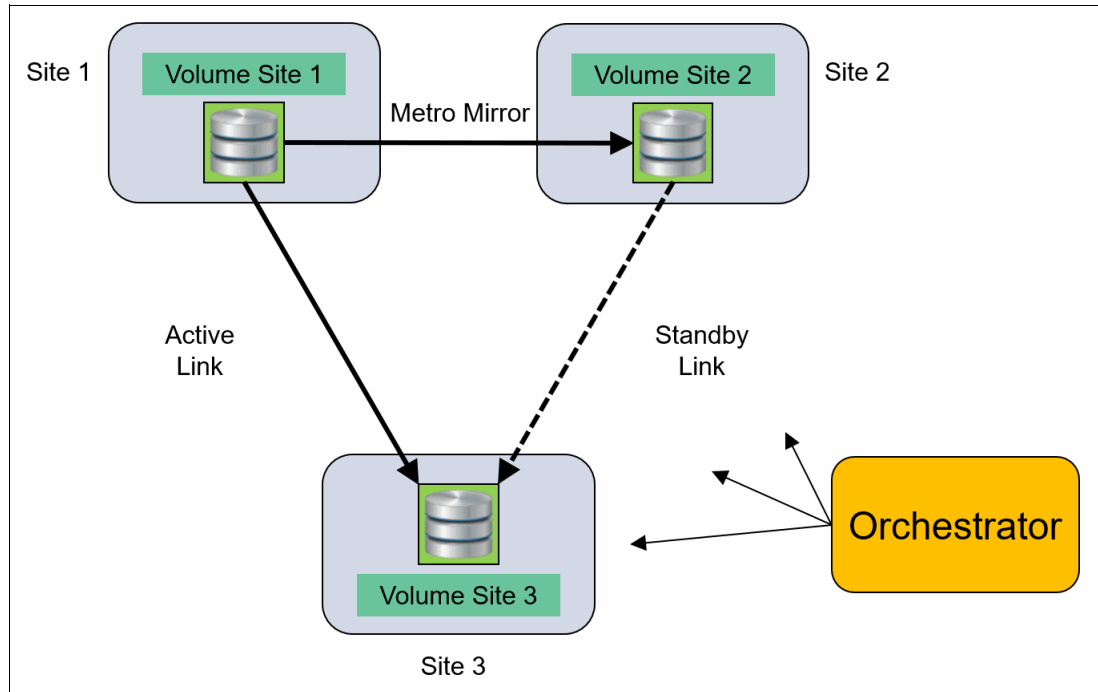


Figure 1-1 3-Site replication overview

The three sites have dedicated functions. The third site always receives data from site one or site two. In normal operation, data will not be replicated from site three to any other site. This may change in a failure scenario. Failure scenarios are described in Chapter 6, “Failure protection and recovery procedures for 3-Site Replication” on page 65.

1.1.1 Terminology

3-Site Replication will use the following terminology for the three Spectrum Virtualize clusters in a 3-Site setup:

- ▶ *Master*: the production site, Site 1 in Figure 1-1
- ▶ *AuxNear*: the near site with a synchronous copy of the production site, Site 2 in Figure 1-1
- ▶ *Near Sites*: the group of the Master site and the AuxNear site.
- ▶ *AuxFar*: the far disaster recovery site, Site 3 in Figure 1-1

The Master site, the AuxNear site, and the AuxFar site will be configured using the 3-Site Replication Orchestrator. In a configuration with only one production site, the Master site will receive the clients production IO workload. A configuration where the Master site and the AuxNear site are receiving IO is also possible. The AuxFar site will always be the site receiving the asynchronously replicated data from the Master, or the AuxNear site.

1.2 Metro Mirror and Remote Copy Consistency Groups

IBM 3-Site Replication uses Metro Mirror and Remote Copy Consistency Groups to mirror and replicate data, and to assure consistency among dependent volumes.

Metro Mirror establishes a synchronous relationship between two volumes of equal size. The volumes in a Metro Mirror relationship are referred to as the *master* volume and the *auxiliary* volume. Traditional Metro Mirror is primarily used in a metropolitan area, or geographical

area, up to a maximum distance of 300 Km (186.4 miles) to provide synchronous replication of data.

The volume properties of master and auxiliary are defined when the synchronous relationship between two volumes is defined. However, the volume serving IO to a host can change. A volume having the role of primary serves IO to a host, and is responsible for replicating data to the secondary volume. The secondary volume is offline and host access is not possible. To access the secondary volume, the roles of the two volumes in a synchronous relationship have to be switched. This implies that host access will only be possible from the new primary volume.

With synchronous copies, host applications write to the master volume, and they receive confirmation that the write operation completed when the data is written to the primary and auxiliary volume. This action ensures that both the volumes have identical data when the write completes. After the initial copy completes, the Metro Mirror function always maintains a fully synchronized copy of the source data at the target site.

Consistency groups can be used to maintain data integrity for dependent writes to multiple volumes. Volumes in a Consistency Group have all the primary volumes on the same site.

Details on Metro Mirror, remote copy Consistency Groups and FlashCopy® are explained in detail in the IBM Redbooks publication:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1*, SG24-8465
- ▶ *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1*, SG24-8466
- ▶ *Implementing the IBM FlashSystem 5010 and FlashSystem 5030 with IBM Spectrum Virtualize V8.3.1*, SG24-8467

3-Site Replication expands the concept of Metro Mirror Consistency Groups with a copy at a far third site.

1.2.1 Topology

The 3-Site Replication is based on Remote Copy Consistency Groups. The volume data of a Consistency Group can be replicated in two different topologies.

- ▶ Star Topology:
 - The primary volumes synchronously mirror data to the auxiliary volumes.
 - The primary volumes asynchronously replicate data to the AuxFar volumes.
 - The primary volume is the source of the replication to the third site.
- ▶ Cascade Topology:
 - The primary volumes synchronously mirror data to the auxiliary volume.
 - The auxiliary volumes asynchronously replicate data to the AuxFar volumes.
 - The auxiliary volume is the source of the replication to the third site.

The source of a data replication for a Consistency Group is either the Master site, or the AuxNear site. A replication is called *active* between a Near Site and the AuxFar site if it is used for replicating data. A replication is called *standby* between a Near Site and the AuxFar site if it is not *active*. The client can switch from a Star to a Cascade Topology, and the reverse is also true. Figure 1-2 shows the two topologies.

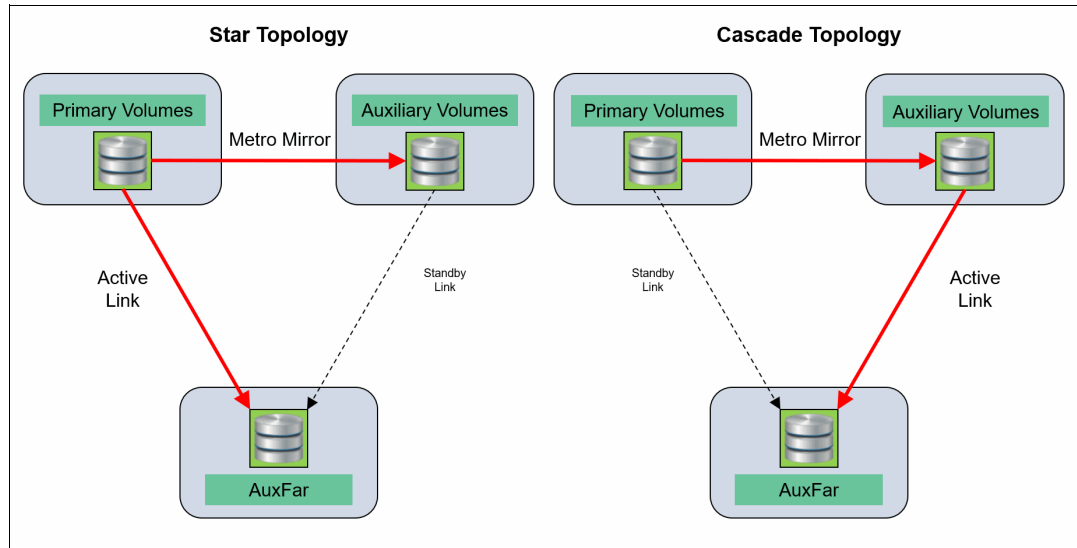


Figure 1-2 Star and Cascade topology

1.3 3-Site Replication: mirroring and replication

3-Site Replication uses Metro Mirror and Consistency Groups to mirror and replicate data and to assure consistency among dependent volumes. 3-Site Replication uses and enhances existing Spectrum Virtualize features and techniques. The basic components used are:

- ▶ Metro Mirror
- ▶ Consistency groups
- ▶ AccessPoint volumes
- ▶ FlashCopy

AccessPoint volumes are dedicated FlashCopy target volumes for mirroring data to the third site.

This section will explain the concepts of data replication and data consistency. This will be shown using only one volume. The 3-Site Consistency Group concept will be discussed in 1.4, “3-Site Replication: Consistency Groups”.

To simplify the description we will assume a star topology, and the primary volume is also the source of the replication to the third site. Also, the primary volume is on the Master site. The cascade topology will be explained later in this section.

A single volume can only participate in one remote copy relationship. The primary volume already has a metro mirror relationship to the auxiliary volume on the other Near Site. For the 3-Site solution, two replications are needed:

1. Synchronously mirrored data to the other Near Site.
2. Periodically asynchronously replicated data to the third site.

The first replication is done by the existing Metro Mirror relationship between the primary and the secondary volume.

The second replication, the replication to the third site, is based on a new set of two volumes, one at the site with the primary volume and one at the AuxFar site. These two new volumes are called AccessPoints and are 3-Site dedicated volumes, and they are in a Metro Mirror

remote relationship. Figure 1-3 shows the primary volume and its corresponding AccessPoint at the Master site. AccessPoints are only accessible for the 3-Site Replication and not for any other use cases. Their behavior is different from standard volumes. This will be explained in this section.

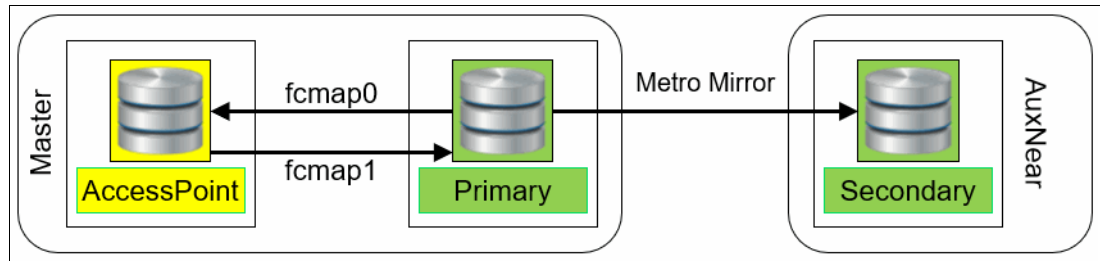


Figure 1-3 Left AccessPoint

AccessPoints of a 3-Site volume exist on the Master, the AuxNear and the AuxFar site. AccessPoints at the Master site are called left AccessPoints. AccessPoints at the AuxNear site are called right AccessPoints.

FlashCopy (FC) mappings are used to ensure consistency of the AccessPoints. Before data is replicated to the third site a FlashCopy is taken from Primary volume. Normal FC behavior ensures the split bitmap is used to read the data from the AccessPoint volume if it has been split (when data was changed on the primary volume), or via a re-directed read to the source volume if not split.

Figure 1-4 shows the Metro Mirror synchronous relationship between the two left AccessPoints, the left AccessPoint of the Primary volume at the Master site, and the left AccessPoints of the third volume (tertiary volume) at the AuxFar site.

Note: The replication between the two AccessPoints is implemented using Metro Mirror replication. The Orchestrator implements periodic cycling by stopping and starting this Metro Mirror to replicate data from a near site to the AuxFar site. Therefore, the data is asynchronously copied to the AuxFar site.

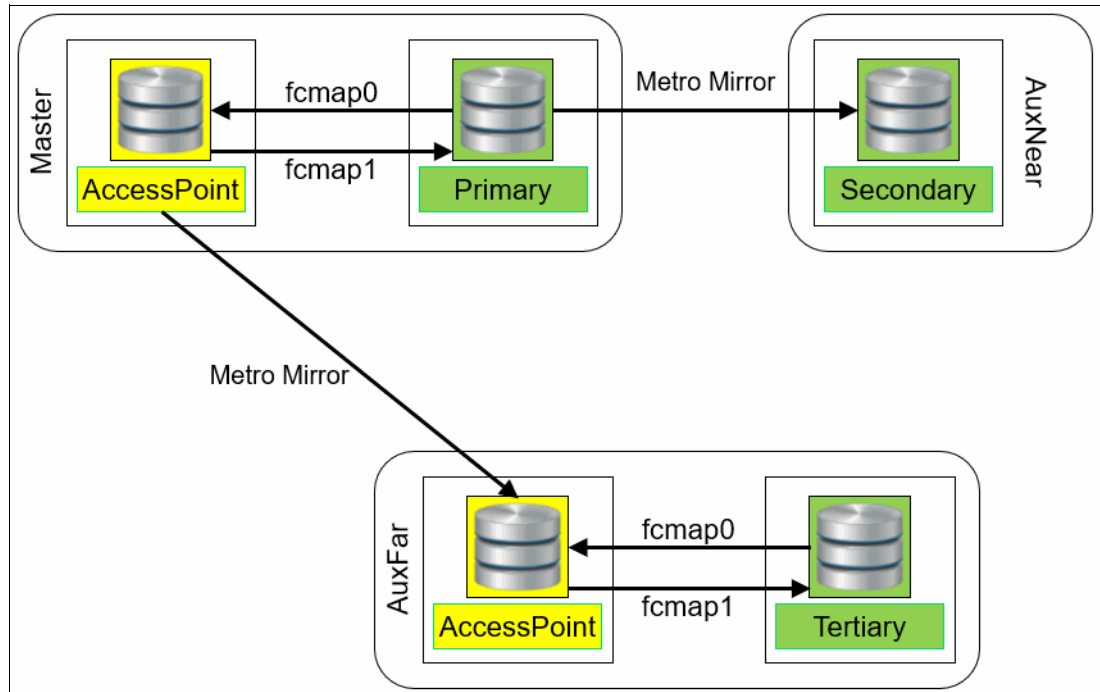


Figure 1-4 Left AccessPoints and their Metro Mirror relationship

Data is replicated periodically from the left AccessPoint to the third site. The initial replication copies all data from the primary volume to the third site. Then, only the changes will be periodically replicated to the third site.

The data replicated is crash consistent meaning that the write order is maintained. First, a FlashCopy of the primary volume is taken, and this assures that all writes before this point in time are included in the FlashCopy. The Metro Mirror relationship between the AccessPoint of the primary volume and the AccessPoint of the tertiary volume transfers the data to the third site.

The Metro Mirror relationship is only used for the transport of the data. Data consistency is assured by the FlashCopy techniques on the Master and AuxFar site. Therefore, the maximum Round Trip Time (RTT) between a Near Site and the AuxFar site is 250 ms, the same as for Global Mirror.

Data at the third site is not written to the AccessPoint volume but directly to the tertiary volume. This behavior is a dedicated 3-Site Replication feature having two advantages:

- The changed blocks of the tertiary volume are copied to its AccessPoint, assuring data consistency in case the transfer is broken.
- There is no need to reverse a FlashCopy. This would only be necessary if the data would be written to the AccessPoint.

Note: The Metro Mirror is defined between the Near Site AccessPoint and the AccessPoint of the tertiary volume. However, data is directly replicated from the Near Site AccessPoint volume to the tertiary volume. No additional reverse copying is needed. This is a dedicated feature for the 3-Site Replication. FlashCopy mappings of the tertiary volumes assure data consistency.

Figure 1-5 shows the 3-Site Replication AccessPoints for the left and the right site.

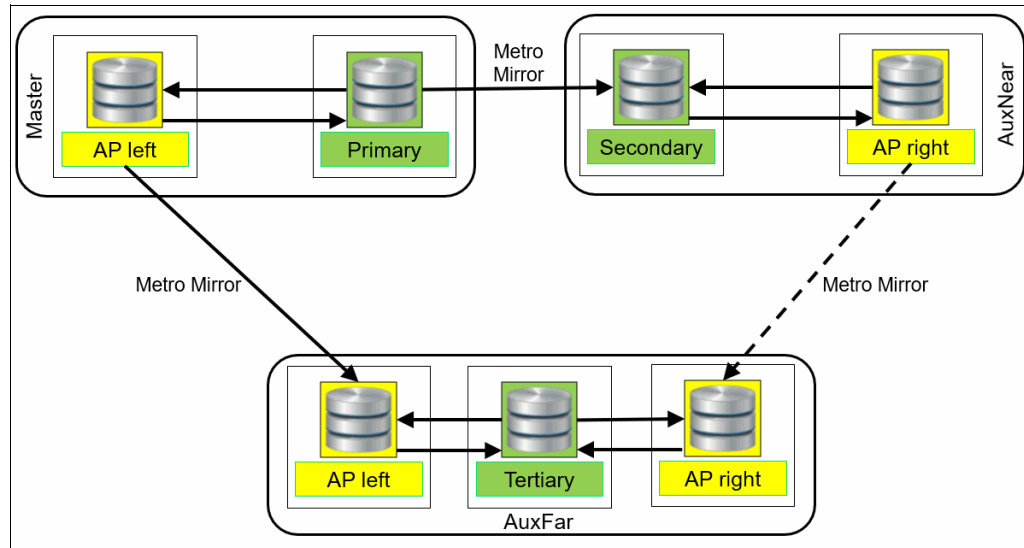


Figure 1-5 3-Site Replication volumes, AccessPoints and FlashCopy mappings

Figure 1-5 shows the Metro Mirror connections between the three sites, the three volumes of the 3-Site Replication, and their AccessPoints. In 1.4, “3-Site Replication: Consistency Groups” we describe the relationships in detail. For a given Consistency Group and all its remote copy relationships, only one AuxFar remote copy will be active to a Near Site. The active Near Site is called the *periodic source*. The left Metro Mirror is shown as solid to indicate that the active remote copy relationships is on the left site, and the Master site is the *periodic source*.

1.4 3-Site Replication: Consistency Groups

3-Site Replication uses Metro Mirror (MM) remote copy relationships (RCs) between every two sites and groups them in remote copy Consistency Groups (CGs).

Figure 1-6 shows a 3-Site configuration having the Consistency Group *CG0*. *CG0* contains two Metro Mirror remote copy relationships to the AuxNear site. The configuration is done using 3-Site Orchestrator commands.

The two Metro Mirror remote copy relationships to the AuxFar are not shown. Only the name of the CG to the third site is shown: *tscgl_nawfbf644*.

The naming for an automatically generated CG follows these rules:

`tscg[lr]_<letters and numbers>`

- ▶ **tscg** is the prefix for all CGs created by the Orchestrator
- ▶ **l** or **r** donates the site of the CG: **left** for the Master to AuxFar CG, and **right** for the AuxNear to AuxFar CG.
- ▶ **letters** and **numbers** is a unique suffix and identical for the left and right CG of the Master to AuxNear CG

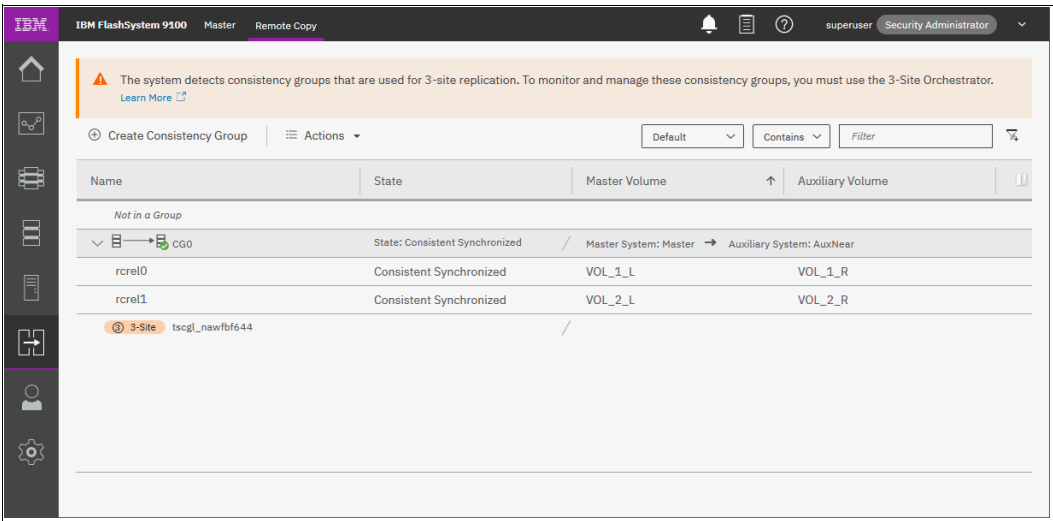


Figure 1-6 Master Remote Copy View

3-Site Orchestrator commands are the only way to create, change, maintain, and remove 3-Site relationships and CGs.

Figure 1-7 shows a 3-Site configuration having a Consistency Group *CG0*. *CG0* contains two Metro Mirror remote copy relationships to the AuxNear site. The configuration is done using the 3-Site Orchestrator commands.

The two Metro Mirror remote copy relationships to the AuxFar are not shown. Only the name of the CG to the third site is shown: *tscgl_nawfbf644*.

Note: To monitor and manage 3-Site Replication Consistency Groups, 3-Site Orchestrator has to be used.

Figure 1-7 shows the AuxNear 3-Site configuration of CG *CG0*. *CG0* contains two Metro Mirror remote copy relationships to the AuxNear site. Like the Master view, the two Metro Mirror remote copy relationships to the AuxFar are not shown. Only the name of the CG to the third site is shown: *tscgr_nawfbf644*. The name only differs in the *r* for right site notation.

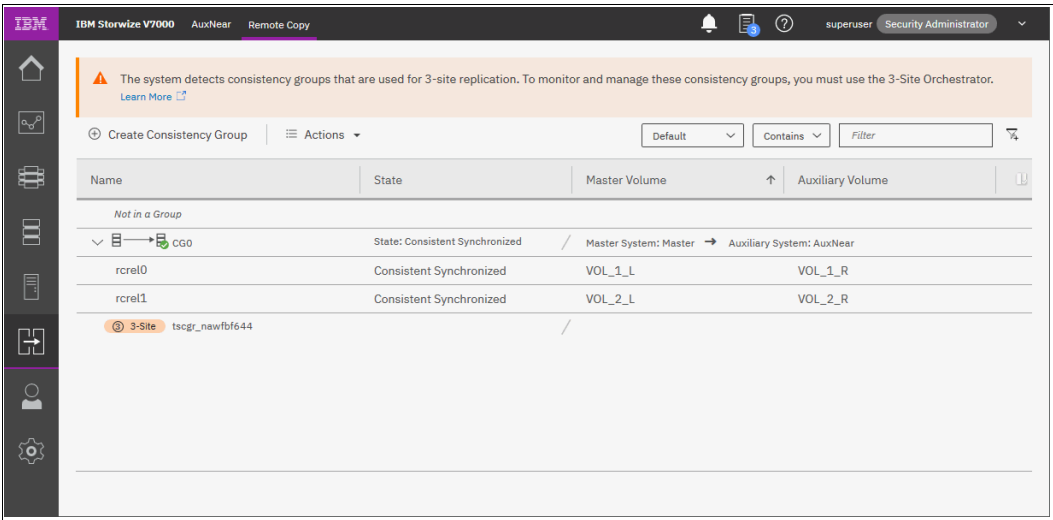


Figure 1-7 AuxNear Remote Copy View

Figure 1-8 shows the AuxFar site. Like the Master and the AuxNear view, only the names for the 3-Site CG to the Master (left) and AuxNear (right) are shown:

- ▶ *tscgl_nawfbf644*: CG between Master and AuxFar.
- ▶ *tscgr_nawfbf644*: CG between AuxNear and AuxFar.

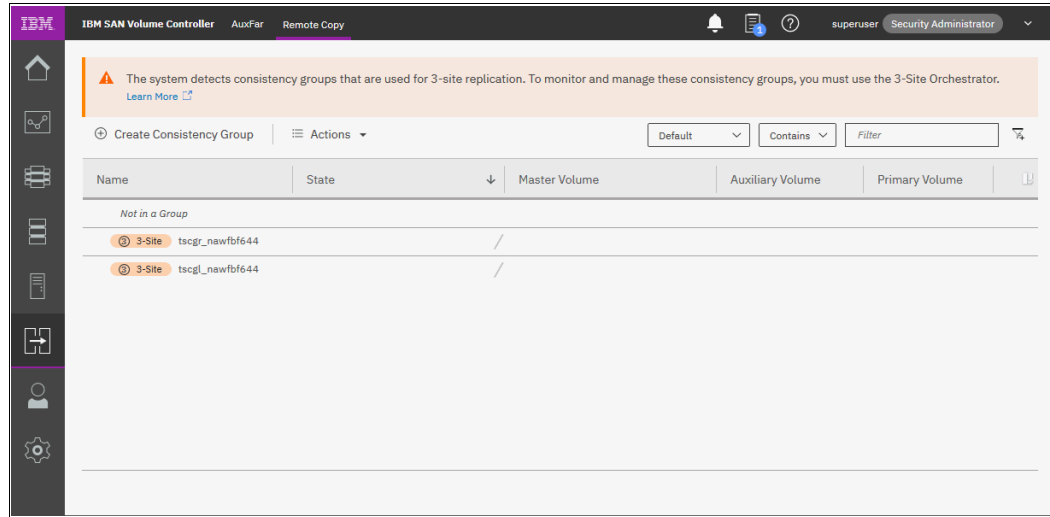


Figure 1-8 AuxFar Remote Copy view

The active relationship can be seen only by using the 3-Site Orchestrator CLI command **ls3sitercconsistgrp** as described in Chapter 5, “3-Site Replication Monitoring and Maintenance” on page 45.

1.4.1 Cluster command line interface details

The Spectrum Virtualize command line interface (CLI) shows the details of the involved volumes, FlashCopy mappings, remote copy relationships, and remote copy Consistency Groups. This section shows the details on every Spectrum Virtualize cluster. The details of a CG shown in the GUI can be listed by the CLI commands shown in the following examples.

The output of the CLI commands, and the header names, are shortened for better readability. Also, only one volume of the CG is shown.

Master details

The volume VOL_1_L has the associated AccessPoint vdisk0. This can be checked using the **lsfcmap** and **lsvdisk** command output as shown in Example 1-1.

Example 1-1 Master volumes and FlashCopy mappings

```
Master:3SiteUser>lsvdisk -delim : | cut -d: -f2,5,13,29 | tr ":" "\t"
name      status  RC_name      function
VOL_1_L   online  rcrc10       master
vdisk0    online  tsrell_ffyvmf72 master

Master:3SiteUser>lsfcmap -delim : | cut -d: -f2,4,6,9,17 | tr ":" "\t"
name      source_vdisk_name target_vdisk_name status      start time
fcmap0    VOL_1_L           vdisk0           copying     200323151004
fcmap1    vdisk0            VOL_1_L           idle_or_copied
```

The CG *tscgl_nawfbf644* as shown in Figure 1-6 can be checked using the **lsrconsistgrp** command as shown in Example 1-2. The Metro Mirror RCs used for the two volumes are listed using the **lsrcrelationship** command as shown in Example 1-2.

Example 1-2 Master RCs and CGs

```
Master:3SiteUser>lsrcrelationship -delim : | cut -d: -f2,4,6,8,10,11,13
name          master_ master_ aux_   aux   primary consistency_group_name
              cluster vdisk  cluster vdisk
rcrel0        Master VOL_1_L AuxNear VOL_1_R master  CG0
tsrell_ffyvmf72 Master vdisk0 AuxFar  vdisk0 master  tscgl_nawfbf644

Master:3SiteUser>lsrconsistgrp -delim : | cut -d: -f2,4,6-8,10 | tr ":" "\t"
name          master_ aux_   primary state                copy_type
              cluster cluster
CG0           Master  AuxNear master  consistent_synchronized metro
tscgl_nawfbf644 Master  AuxFar  master  consistent_stopped      metro
```

AuxNear details

The volume VOL_1_R has the associated AccessPoint vdisk0. This can be checked using the **lsfcmap** and **lsvdisk** command output as shown in Example 1-3. AuxNear volumes and FlashCopy mappings

Example 1-3 AuxNear RCs and CGs

```
AuxNear:3SiteUser>lsvdisk -delim : | cut -d: -f2,5,13,29 | tr ":" "\t"
name  status RC_name      function
VOL_1_R online  rcrel0      master
vdisk0 online  tsrellr_ffyvmf72 master

AuxNear:3SiteUser>lsfcmap -delim : | cut -d: -f2,4,6,9,17 | tr ":" "\t"
name  source_vdisk_name target_vdisk_name status      start_time
fcmap0 VOL_1_R          vdisk0           copying     200323151004
fcmap1 vdisk0           VOL_1_R          idle_or_copied
```

The CG *tscgr_nawfbf644*, as shown in Figure 1-7, can be checked using the **lsrconsistgrp** command as shown in Example 1-4. The Metro Mirror relationships used for the two volumes are listed using the **lsrcrelationship** command as shown in Example 1-4. The RC name and the CG to the AuxFar site of the corresponding volumes on the AuxNear cluster have the same prefix and suffix. They differ only in the site notation, **l** for the Master, and **r** for the AuxNear site.

Example 1-4 AuxNear RCs and CGs

```
AuxNear:3SiteUser>lsrcrelationship -delim : | cut -d: -f2,4,6,8,10,11,13
name          master_ master_ aux_   aux   primary consistency_group_name
              cluster vdisk  cluster vdisk
rcrel0        Master VOL_1_L AuxNear VOL_1_R master  CG0
tsrellr_ffyvmf72 AuxNear vdisk0 AuxFar  vdisk1      tscgr_nawfbf644

AuxNear:3SiteUser>lsrconsistgrp -delim : | cut -d: -f2,4,6-8,10 | tr ":" "\t"
name          master_ aux_   primary state                copy_type
              cluster cluster
CG0           Master  AuxNear master  consistent_synchronized metro
```

tscgr_nawfbf644 AuxNear AuxFar

idling

metro

AuxFar details

The volume VOL_1_T has two associated AccessPoint, vdisk0 and vdisk1. One will be used for the RC to the Master cluster (left site), and the one for the RC to the AuxNear cluster (right site). Only one RC will be active depending on the periodic source.

This can be checked using the **lsfcmap** and **lsvdisk** command output as shown in Example 1-5. If the RC of the left or of the right site is active, they can be listed using the *channel* property of the RC associated with the active volume. In Example 1-5 the volume *vdisk0* is online and therefore the RC *tsrell_ffyvmf72* is active. The *channel* property of the RC *tsrell_ffyvmf72* is left.

Example 1-5 AuxFar volumes and FlashCopy mappings

```
AuxFar:3SiteUser>lsvdisk -delim : | cut -d: -f2,5,13,29 | tr ":" "\t"
name      status  RC_name function
VOL_1_T   online
vdisk0     online  tsrell_ffyvmf72 aux
vdisk1     offline tsreir_ffyvmf72 aux

:AuxFar:3SiteUser>lsfcmap -delim : | cut -d: -f2,4,6,9,17 | tr ":" "\t"
name      source_vdisk_name  target_vdisk_name  status  start_time
fcmap0    VOL_1_T vdisk0  copying 200323183024
fcmap1    vdisk0 VOL_1_T idle_or_copied
fcmap2    VOL_1_T vdisk1 idle_or_copied
fcmap3    vdisk1 VOL_1_T idle_or_copied

AuxFar:3SiteUser>lsrcrelationship tsrell_ffyvmf72 | grep channel
channel left
AuxFar:3SiteUser>lsrcrelationship tsreir_ffyvmf72 | grep channel
channel right
```

The CGs *tscgl_nawfbf644* and *tscgr_nawfbf644* (left and right CGs), as shown in Figure 1-8, can be checked using the **lsrconsistgrp** command as shown in Example 1-6. The Metro Mirror RCs used for the two volumes are listed using the **lsrcrelationship** command as shown in Example 1-6. The RC name and the CG to the Near Sites of the corresponding volumes on the Master cluster have the same prefix and suffix. They differ only in the site donation, **l** for the Master (left) and **r** for the AuxNear (right) site.

Example 1-6 AuxFar RCs and CGs

```
Master:3SiteUser>lsrcrelationship -delim : | cut -d: -f2,4,6,8,10,11,13
name      master_ master_ aux_  aux  primary consistency_group_name
           cluster vdisk cluster vdisk
tsrell_ffyvmf72 Master vdisk0 AuxFar vdisk0 master tscgl_nawfbf644
tsreir_ffyvmf72 AuxNear vdisk0 AuxFar vdisk1          tscgr_nawfbf644

Master:3SiteUser>lsrconsistgrp -delim : | cut -d: -f2,4,6-8,10 | tr ":" "\t"
name      master_ aux_  primary state  copy_type
           cluster cluster
AuxFar:3SiteUser>lsrconsistgrp -delim : | cut -d: -f2,4,6-8,10 | tr ":" "\t"
name      master_cluster_name  aux_cluster_name  primary state  copy_type
tscgl_nawfbf644 Master AuxFar master consistent_stopped metro
```


1.5 Data consistency

The Metro Mirror relationship between a Near Site and the AuxFar site is only used for the transport of the data. Data consistency is assured by FlashCopy on the Near Site and AuxFar site.

At the Near Site, the FlashCopy of the primary volume and the secondary volume is taken at the same point in time to ensure that Flash Copies of both Near Sites contain the same data. Therefore, the AccessPoints of the primary and secondary volume contain identical data and they are crash consistent, meaning that the write order is maintained.

New and changed data is periodically mirrored from a Near Site AccessPoint volume to the tertiary volume using the Metro Mirror relationship. The FlashCopy split bitmap is used to read the data from the AccessPoint volume if it has been split (when data was changed on the primary volume), or via a re-directed read to the source volume if not split (when data was not changed on the primary volume). Copy on write techniques maintain the previous data of the tertiary volume in the associated AccessPoint. If needed, a roll-back to get the state before the mirroring started is always possible. When all data from a Near Site is replicated, then the Metro Mirror is stopped and the FlashCopy from the tertiary volume to its AccessPoint is restarted.

The data mirroring directly to the tertiary volume, but not to its AccessPoint, is a dedicated feature of the 3-Site Replication.

Note: The Metro Mirror is defined between the Near Site AccessPoint and the AccessPoint of the tertiary volume. However, data is directly replicated from the Near Site AccessPoint volume to the tertiary volume. No additional reverse copying is needed. This is a dedicated feature for the 3-Site Replication. FlashCopy mappings of the tertiary volumes assure data consistency.

Figure 1-9 shows a high level example on periodic Metro Mirror data replication from a Near Site to AuxFar. At time stamps T2 and T3, the Metro Mirror is active and transferring the data. The figure shows the data content of the primary and tertiary volume at different points in time. To the left to the volume icon, the date and its time stamps are shown.

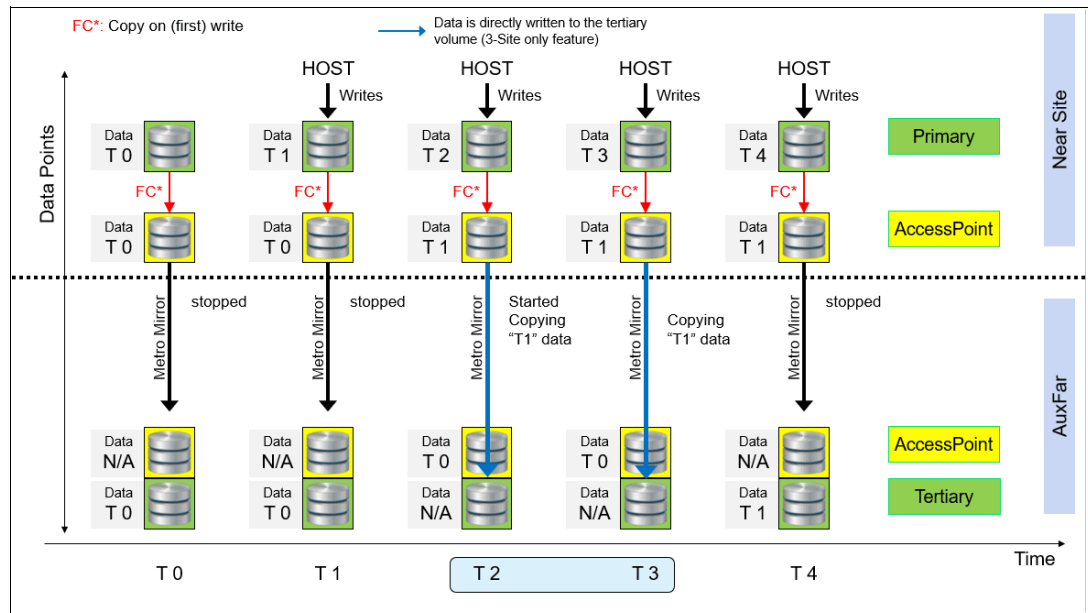


Figure 1-9 Cycling Idea

The content of the volumes and the status of the Metro Mirror at the different points in time are:

- ▶ Time stamp T0:
 - All data was transferred to the third site using Metro Mirror.
 - The FlashCopy “Primary → AccessPoint” is running (status “copying”).
 - Metro Mirror is stopped (state “consistent_stopped”) and the changed AccessPoint blocks are recorded, which have to be transferred to the third site.
- ▶ Time stamp T1:
 - The host writes data to the primary volume.
 - The AccessPoint keeps the changed blocks using “copy on (first) write”.
 - Metro Mirror keeps the information on the changed blocks.
- ▶ Time stamp T2:
 - A new primary volume FlashCopy is taken at T2, and the Metro Mirror to AuxFar is started. The FlashCopy behavior assures data consistency.

A host write is either stored in the previous FlashCopy, or in the new FlashCopy.

 - The Metro Mirror replication that has started to AuxFar is in the state *inconsistent_copying*, and is transferring the changes between T0 and T2 to the third site. Normal FlashCopy behavior ensures that the data is read either from the primary volume, or from the data from the AccessPoint if it has changed.

Although the Metro Mirror relationship is defined between the two AccessPoints, the data is directly written to the tertiary volume, and the changed blocks are saved in the AccessPoint volume for data consistency. This is a dedicated feature of 3-Site Replication.
- ▶ Time stamp T3:
 - The host writes data to the primary volume.
 - AccessPoint contains changes (using “copy on write”).

- ▶ Time stamp T4:
 - Metro Mirror has transferred all data, and is the gap between T0 and T2 to the third site.
 - Metro Mirror is stopped (state “consistent_stopped”).
 - This is equivalent to time stamp T0.

1.6 3-Site user

3-Site Replication commands can only be executed by a user having the 3-Site administrator role. 3-Site Orchestrator will use a user that has the 3-Site administrator role to connect to the three clusters, and to execute commands. The 3-Site administrator role is described in Chapter 3, “Configuring a 3-Site Replication solution” on page 23.



Planning for 3-Site Replication

This chapter describes the steps that are required to plan the installation and configuration of 3-Site replication in your environment. It includes inter-site link and partnership requirements, 3-Site Orchestrator requirements, partnership planning, capacity planning for access points, general recommendations, and restrictions.

Note: Make sure that the planned configuration is reviewed by IBM or an IBM Business Partner before implementation. Such a review can increase both the quality of the final solution and prevent configuration errors that could impact solution delivery.

This chapter includes the following topics:

- ▶ Inter-site link and partnership requirements
- ▶ 3-Site Orchestrator requirements
- ▶ AccessPoint capacity planning and copy on (first) write performance impact
- ▶ General requirements
- ▶ Restrictions and limitations

2.1 Inter-site link and partnership requirements

In order to plan for 3-Site Replication, consider that the only partnership method supported at the time of writing is Fibre Channel (FC). The FC partnership must be in place between the three systems from each site.

Consider the following points when planning your partnerships:

- ▶ At the time of writing, only FC partnerships are supported
- ▶ Assess your link requirements and plan for your Metro Mirror relationship
 - Consider the number of I/O groups your system will have
 - The maximum supported RTT between the systems in the fibre channel relationship is the same supported in Global Mirror, 250ms
- ▶ Calculate your inter-site bandwidth link

The two major parameters of a link are its bandwidth and latency. Latency might limit the maximum bandwidth available over IP links depending on the details of the technology used, like the circuit type (such as sonet ring), SAN hardware and connections (such as FCIP routers, transceivers or ISL).

When planning the inter-cluster link, take into account the peak performance that is required. This consideration is especially important for Metro Mirror configurations.

When Metro Mirror is used a certain amount of bandwidth is required for the system inter-cluster heartbeat traffic. The amount of traffic depends on how many nodes are in each of the two clustered systems.

Table 2-1 shows the amount of heartbeat traffic, in megabits per second, that is generated by various sizes of clustered systems.

Table 2-1 Intersystem heartbeat traffic in Mbps

SAN Volume Controller System 1	SAN Volume Controller System 2			
	2 nodes	4 nodes	6 nodes	8 nodes
2 nodes	5	6	6	6
4 nodes	6	10	11	12
6 nodes	6	11	16	17
8 nodes	6	12	17	21

Note: Table 2-1 uses the SAN Volume Controller as an example, but you can use the same values based on your FlashSystem controllers amount.

These numbers estimate the amount of traffic between the two clustered systems when no I/O is taking place to mirrored volumes. Half of the data is sent by each of the systems. The traffic is divided evenly over all available inter-cluster links. Therefore, if you have two redundant links, half of this traffic is sent over each link.

The bandwidth between sites must be sized to meet the peak workload requirements. You can estimate the peak workload requirement by measuring the maximum write workload averaged over a period of 1 minute or less, and adding the heartbeat bandwidth. Statistics must be gathered over a typical application I/O workload cycle, which might be days, weeks, or months, depending on the environment on which the system is used.

When planning the inter-site link, consider also the initial sync and any future resync workloads. It might be worthwhile to secure additional link bandwidth for the initial data synchronization.

If the link between the sites is configured with redundancy so that it can tolerate single failures, you must size the link so that the bandwidth and latency requirements are met even during single failure conditions.

When planning the inter-site link, make a careful note whether it is dedicated to the inter-cluster traffic or if it is going to be used to carry any other data. Sharing the link with other traffic (for example, cross-site IP traffic) might reduce the cost of creating the inter-site connection and improve link utilization. However, doing so might also affect the links' ability to provide the required bandwidth for data replication.

Note: Contact your IBM Sales representative or IBM Business partner to perform these calculations.

2.2 3-Site Orchestrator requirements

When planning for the 3-Site Orchestrator installation, you need to deploy a server that meets the following hardware and software requirements:

- ▶ Red Hat Enterprise Linux (RHEL) version 7.5 or later. Then install 3-Site Orchestrator on an x64-bit architecture server.
- ▶ The server can be physical or virtual.
- ▶ The server must have a minimum of 4GB random access memory (RAM).
- ▶ The server must have a minimum 4-core microprocessors.
- ▶ The server must have 300 MB free storage space for RPM file and installation.
- ▶ IP network access to all three sites must be available.
- ▶ The OpenSSH utility must be installed.
- ▶ Passwordless SSH login must be available for all three sites from the 3-Site Orchestrator host.
- ▶ The physical location of the host can be at any of the three sites or at a fourth site.
- ▶ Firewall requirements: three firewall ports must be open for SSH tunneling (for example, 6000-6002) and port 7003 must also be open for inter process communication between the Orchestrator Daemon and the Event Monitor Daemon.
- ▶ All three Spectrum Virtualize systems must be running at least V8.3.1.1 code.
- ▶ All three Spectrum Virtualize systems must have Fibre Channel remote copy partnerships and fulfill remote copy license requirements.

2.3 AccessPoint capacity planning and copy on (first) write performance impact

As part of 3-Site replication configuration, the 3-Site Orchestrator will create dedicated volumes that handle asynchronous data replication among the three sites, and these volumes are referred to as AccessPoints. Not all accesspoint volumes are intended to be used at the same time. The accesspoint volumes for primary and secondary volumes, those located on near sites, will be used all the time to have a consistent image of primary and secondary volumes. If you are using a star topology, left accesspoint volume from tertiary volume will be

writing data, and if you are using cascade topology, right accesspoint volume from tertiary volume will be writing data.

Note: Right accesspoint volumes from the tertiary volumes will be used on the star topology only in a case of a link failure between the Master and AuxFar site. Left accesspoint volumes from the tertiary volumes will be used on the cascade topology only in a case of link failure between AuxNear and AuxFar sites.

For capacity planning purposes consider the following:

- ▶ 3-Site Orchestrator will create all accesspoint volumes, you cannot create them manually.
- ▶ 3-Site Orchestrator will use the same storage pool of a volume to create its accesspoint volume.
- ▶ If the volume has a volume copy, 3-Site Orchestrator will use the same pool as the primary copy of the volume.
- ▶ 3-Site Orchestrator will create the accesspoint volume as compressed if it is in a Data Reduction Pool (DRP) containing arrays with compressing drives.
- ▶ 3-Site Orchestrator will create the accesspoint volume as thin with auto-expand on if in a DRP containing arrays with regular drives.
- ▶ 3-Site Orchestrator will create the accesspoint volume as thin with auto-expand on if in a standard pool.
- ▶ It is not possible to change the accesspoint volume settings.

Note: The accesspoint volumes will normally consume capacity equal to the initially specified rsize 2%. During re-synchronization, the accesspoint volumes will grow as they retain the changed data, so it will depend on how actively your data is overwritten. If your data changes 30% then the accesspoint volumes will grow 30%. After re-synchronization the accesspoint volume will shrink back to their initially-specified rsize 2%.

Performance impacts due to “copy on (first) write”

FlashCopy uses copy on (first) write to keep a consistent copy of the data in the accesspoint. This might increase latency on very high write loads when the write cache gets filled up due to slow backend storage.

This is more likely to happen in cases where you have spinning drive hardware, or SAN Volume Controller systems with slow backend storage. Plan carefully, and assess your storage performance (mostly cache utilization) using IBM Spectrum Control or IBM Storage Insights.

2.4 General requirements

This section intends to provide general recommendations for your 3-Site replication environment planning:

- ▶ Ensure that you have the time zone correctly configured on all three systems and the 3-Site Orchestrator server. This will prevent timestamp inconsistencies and issues to your environment.
- ▶ Have a *Network Time Protocol* (NTP) server configured on all three systems and the 3-Site Orchestrator server, and this will prevent timestamp inconsistencies and issues to your environment.
- ▶ Use only 3-Site Orchestrator to manage your 3-Site objects.
- ▶ All three systems need to have a Remote Copy license.

2.5 Restrictions and limitations

When planning for your 3-Site Replication, consider the restrictions and limits in place at the time of writing:

- ▶ 3-Site Replication does not support HyperSwap® between the two near sites.
- ▶ 3-Site Replication is not supported over an IP partnership
- ▶ Standard Metro Mirror restrictions apply for the near site.
- ▶ Standard Global mirror restrictions apply for the far site.
- ▶ The GUI does not support setting up, managing and monitoring 3-Site Replication.
- ▶ Only one 3-Site Orchestrator host instance can be active at a time.
- ▶ Maximum of 16 Consistency Groups supported with up to 256 relationships in each Consistency Group.
 - There is an overall limit of relationships which is dependent on platform. For platforms that support 10k volumes, the limit is 1250 3-Site relationships. For platforms that support 8192 volumes it is 1024
- ▶ Standalone relationships are not supported in 3-Site.
- ▶ Minimum cycle period for far site replication is 5 minutes, and RPO is 10 minutes.
- ▶ Volume resizing is not supported as they are part of FlashCopy relationships.
- ▶ VVOLs are not supported on 3-Site Replication as VVOLs do not support remote copy relationships.
- ▶ Metro Mirror consistency protection feature is not supported in 3-Site Replication configurations.



Configuring a 3-Site Replication solution

This chapter provides instructions on the initial configuration of a 3-Site replication solution. It includes installing and setting up the 3-Site Orchestrator, preparing Spectrum Virtualize clusters, and creating a 3-Site configuration.

Steps given in this chapter assume that Spectrum Virtualize systems are already set up, that hardware is installed and a basic configuration using the Setup Wizard was performed.

This chapter requires the reader to have a knowledge of Spectrum Virtualize copy services. For more details on how they operate, refer to:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1, SG24-8465*
- ▶ *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1, SG24-8466*

This chapter contains the following sections:

- ▶ Preparing the 3-Site Orchestrator host
- ▶ Preparing Spectrum Virtualize clusters
- ▶ Creating a 3-Site configuration
- ▶ After the 3-Site configuration is created, you can start creating 3-Site replication.

3.1 Preparing the 3-Site Orchestrator host

The steps listed below will prepare a Linux host for operation as a 3-Site Orchestrator.

1. Install Orchestrator software

- a. Obtain the 3-Site Orchestrator '.rpm' file from IBM and upload it to your Linux host.
- b. Install it using the **rpm** command as shown in Example 3-1.

Example 3-1 Installing orchestrator

```
[root@localhost ~]# rpm -i ibm-SpectrumVirtualize-rc3site-orchestrator-1...rpm
[root@localhost ~]# rpm -qa | grep 3site
ibm-SpectrumVirtualize-rc3site-orchestrator-1...x86_64
```

- c. 3-Site Orchestrator components are installed to `/opt/ibm/SpectrumVirtualize/rc3site/` directory. Also, three services are installed:
 - `/etc/systemd/system/rc3site.eventd.service` - communicates with all three sites and monitors important attributes of the sites and configured Consistency Groups.
 - `/etc/systemd/system/rc3site.tsmiscservicesd.service` - maintains an audit log of all 3-Site Orchestrator commands.
 - `/etc/systemd/system/rc3site.ttaskd.service` - controls replication on configured 3-Site Orchestrator Consistency Groups.

The host administrator can use the **systemctl** command to make sure that the service is running as shown in Example 3-2

Example 3-2 TaskDaemon status check

```
[root@localhost ~]# systemctl status rc3site.ttaskd.service
? rc3site.ttaskd.service - rc3site ttaskd service. Orchestrator VERSION =
1.0.200309115013
   Loaded: loaded (/etc/systemd/system/rc3site.ttaskd.service; enabled; vendor
   preset: disabled)
   Active: active (running) since Mon 2020-03-23 17:38:11 UTC; 11h ago
   Process: 12140 ExecStart=/usr/bin/ttaskd start (code=exited,
   status=0/SUCCESS)
  Main PID: 12144 (ttaskd)
    CGroup: /system.slice/rc3site.ttaskd.service
            L-12144 /usr/bin/ttaskd start
```

- d. After 3-Site Orchestrator is installed, 3-Site configuration commands become available.
- ### 2. Prepare passwordless communication with Spectrum Virtualize systems.
- a. Ensure that SSH communication is possible from your 3-Site Orchestrator Linux host to Spectrum Virtualize systems on all three replication sites.
 - b. Make sure that RSA signatures for the cluster IP addresses of all three Spectrum Virtualize systems are present in `/root/.ssh/known_hosts`, or add them there, as shown in Example 3-3.

Example 3-3 Adding signatures to known_hosts

```
[root@localhost ~]# ssh-keyscan <ReplMasterIP> >> /root/.ssh/known_hosts
[root@localhost ~]# ssh-keyscan <ReplAuxNearIP> >> /root/.ssh/known_hosts
[root@localhost ~]# ssh-keyscan <ReplAuxFarIP> >> /root/.ssh/known_hosts
```

- c. Generate the SSH key pair for passwordless communication using the **ssh-keygen** command, as shown in Example 3-4. If no parameters are given, **ssh-keygen** generates 2048 bits RSA keys. Do not specify a passphrase.

Example 3-4 Generating key pair for passwordless communication

```
[root@localhost ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa): /root/.ssh/3site
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/3site.
Your public key has been saved in /root/.ssh/3site.pub.
The key fingerprint is:
SHA256:765KxaRLJ3SYQQIcbLqapsuFrpm6xv2kSmqyUuqKXD0 root@localhost
The key's randomart image is:
+---[RSA 2048]-----+
...
+----[SHA256]-----+
```

- d. Copy the public SSH key to /tmp directory on each Spectrum Virtualize system with scp, as Example 3-5 shows. If you plan to use the GUI to set up a user id on Spectrum Virtualize systems, download keys from the Orchestrator host to your PC.

Further steps are needed to configure passwordless authentication are performed from Spectrum Virtualize cluster GUI or CLI and are covered in the next section.

Example 3-5 Copying public key

```
[root@localhost ~]# scp /root/.ssh/3site.pub superuser@<ReplMasterIP>:/tmp/
Password:
3site.pub                                100% 392      1.3MB/s   00:00
[root@localhost ~]# scp /root/.ssh/3site.pub superuser@<ReplAuxNearIP>:/tmp/
Password:
3site.pub                                100% 392      1.3MB/s   00:00
[root@localhost ~]# scp /root/.ssh/3site.pub superuser@<ReplAuxFarIP>:/tmp/
Password:
3site.pub                                100% 392      1.3MB/s   00:00
```

3.2 Preparing Spectrum Virtualize clusters

Perform the following steps to prepare the Spectrum Virtualize clusters at Master, AuxNear and AuxFar sites to work in a 3-Site replication solution.

1. Verify that all three clusters are configured with unique system names. The system name is set during system initialization. It is displayed in the top line of a GUI page, and it is also shown in a CLI prompt.

If required, the system name can be changed using the **chsystem -name <new_system_name>** command.

Note: The iSCSI IQN for each node is generated by using the system and node names. If you are using the iSCSI protocol, changing either name also changes the IQN of all of the nodes in the system and might require reconfiguration of all iSCSI-attached hosts.

System names must remain different through the life of the 3-Site configuration.

2. Check that the Remote Mirroring feature is licensed on all three clusters. To do this, navigate to **Settings** → **System** → **Licensed Functions** in the GUI or use the `lslicense` command. If needed, correct the licensed capacity (or the number of enclosures) to match it to the purchased licensed capacity.
3. Create replication partnerships between the systems. There must be a partnership configured from each system to the other two, so all of them are inter-connected, as illustrated in Figure 3-1.

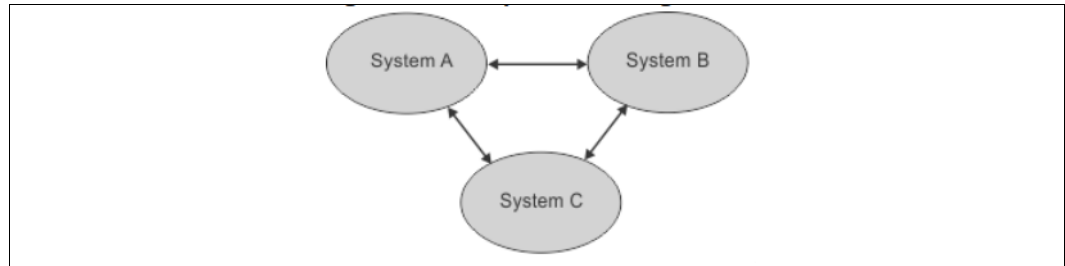


Figure 3-1 Partnership scheme

To create partnerships, you may use the CLI or the GUI. These sections only provide a brief overview of the process, for more details refer to:

Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1, SG24-8465

Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1, SG24-8466

Note: At the time of writing, only FC partnerships are supported for 3-Site replication. IP partnerships are not supported.

- To configure partnerships between clusters using the CLI, perform the following steps:
 - i. Connect to the CLI of system A in Figure 3-1 (for example, Master site). Run the `lspartnershipcandidate` command to list the clustered systems available for setting up a partnership with the local system, and use the `mkfcpartnership` command to create partnerships with systems B and C, as shown in Example 3-6. Specify the bandwidth available for replication in Mbits per second and the maximum bandwidth percentage that can be used for background copy.

Example 3-6 Listing available partners and creating partnerships

```

IBM_Spectrum_Virtualize:ReplMaster:superuser>lspartnershipcandidate
id                configured name
0000001C6EE00020 no      ReplAuxNear
0000001C65200018 no      ReplAuxFar
IBM_Spectrum_Virtualize:ReplMaster:superuser>mkfcpartnership
-linkbandwidthmbits 32000 -backgroundcopyrate 50 ReplAuxNear
IBM_Spectrum_Virtualize:ReplMaster:superuser>mkfcpartnership
-linkbandwidthmbits 32000 -backgroundcopyrate 50 ReplAuxFar
IBM_Spectrum_Virtualize:ReplMaster:superuser>lspartnership
id                name          location partnership      type
0000001C61200002 ReplMaster  local
0000001C6EE00020 ReplAuxNear remote  partially_configured_local fc
0000001C65200018 ReplAuxFar  remote  partially_configured_local fc
  
```

- ii. The partnership is created in *partially_configured_local* state. To get it to *fully_configured*, **mkfcpartnerhsip** needs to be run on system B and system C (ReplAuxNear and ReplAuxFar) against a cluster id or name of system A, as shown in Example 3-7 on page 27.

Example 3-7 Making a partnership fully_configured

```
IBM_Spectrum_Virtualize:ReplAuxNear:superuser>lspartnershipcandidate
id              configured name
0000001C61200002 yes      ReplMaster
0000001C65200018 no       ReplAuxFar
IBM_Spectrum_Virtualize:ReplAuxNear:superuser>mkfcpartnership
-linkbandwidthmbits 32000 -backgroundcopyrate 50 ReplMaster
IBM_Spectrum_Virtualize:ReplAuxNear:superuser>lspartnership
id              name              location partnership      type
0000001C6EE00020 ReplAuxNear local
0000001C61200002 ReplMaster  remote   fully_configured fc
```

- iii. As the last step, the partnership between system B and system C needs to be created in the same way, by running **mkfcpartnership** on both systems.
 - iv. Verify that **lspartnership** on each system shows two remote clusters with a partnership in *fully_configured* state.
- To configure partnerships using the GUI, the following steps must be done:
- i. Connect to the GUI of system A in Figure 3-1 (for example, Master site), navigate to **Copy Services** → **Partnerships** and click on the **Create Partnership** button. After you select the FC partnership type the dialog shown in Figure 3-2 appears. In the drop-down menu, select the partner system (for example, on a AuxNear site), specify the link bandwidth and background copy rate, and click Create.

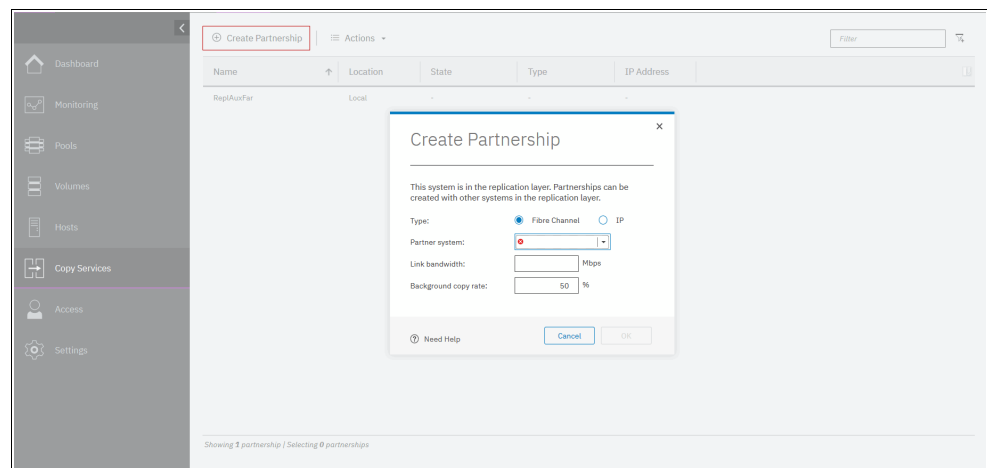


Figure 3-2 Create Partnership dialog

- ii. Repeat the previous step to create a partnership with the AuxFar system. The GUI will show two partnerships in *partially_configured_local* state.
- iii. To get the partnerships to a *fully_configured* state, go to the GUI of each Aux system and perform the same steps to create a partnership with the Master cluster.
- iv. As the last step, the partnership between system B and system C (AuxNear and AuxFar) needs to be created in the same way.
- v. As the result, each pair of sites must be in a *fully_configured* partnership.

4. Configure a user with a 3-Site Administrator role on each of three Spectrum Virtualize systems. It is used by 3-Site Orchestrator to track and control replication processes.

The 3-Site Administrator user must belong to the 3-Site Administrator user group, which does not exist on the system by default, and needs to be created.

- To configure a 3-Site Administrator user using the CLI, run the following commands:

- i. Create a user group with a descriptive name and *3SiteAdmin* role by using **mkusergrp** command, as in Example 3-8.

Example 3-8 Creating a user group

```
IBM_Spectrum_Virtualize:Rep1Master:superuser>mkusergrp -name 3SiteAdmin
-role 3SiteAdmin
User Group, id [6], successfully created
```

- ii. Create a local user record for 3-Site Orchestrator, and specify the SSH key file that was copied to the system as described in section d on page 25. When you run the command, the SSH key is copied into system state and activated for the user, and the input file is deleted. Example 3-9 shows how the **mkuser** command is used for this task.

Example 3-9 Creating 3site orchestrator user

```
IBM_Spectrum_Virtualize:Rep1Master:superuser>mkuser -name 3SiteOrch
-usergrp 3SiteAdmin -keyfile /tmp/3site.pub
User, id [1], successfully created
```

- iii. Execute steps i and ii on other two clusters. Note that the user name specified on all three clusters must match, and must use the same SSH public key.
- If you prefer configuring users with the GUI, perform the following:
 - i. Navigate to **Access** → **Users by group** and click on **Create User Group** button, as shown in Figure 3-3.

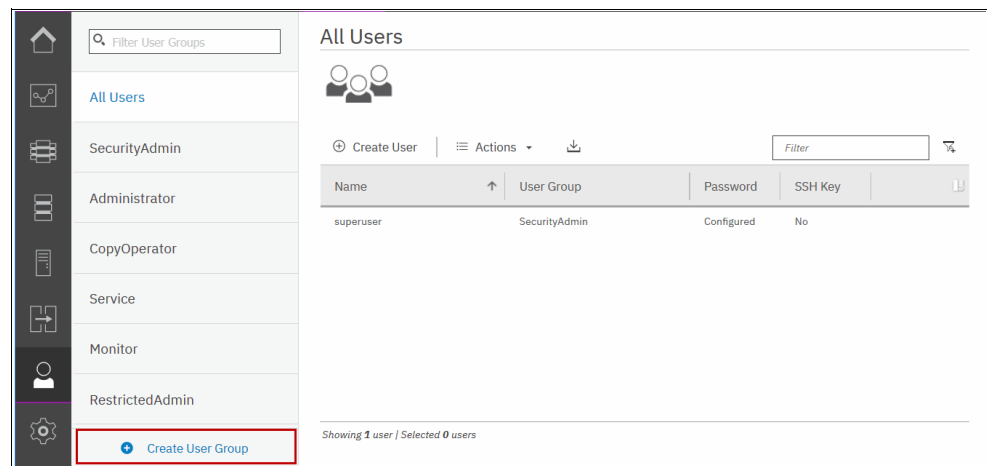
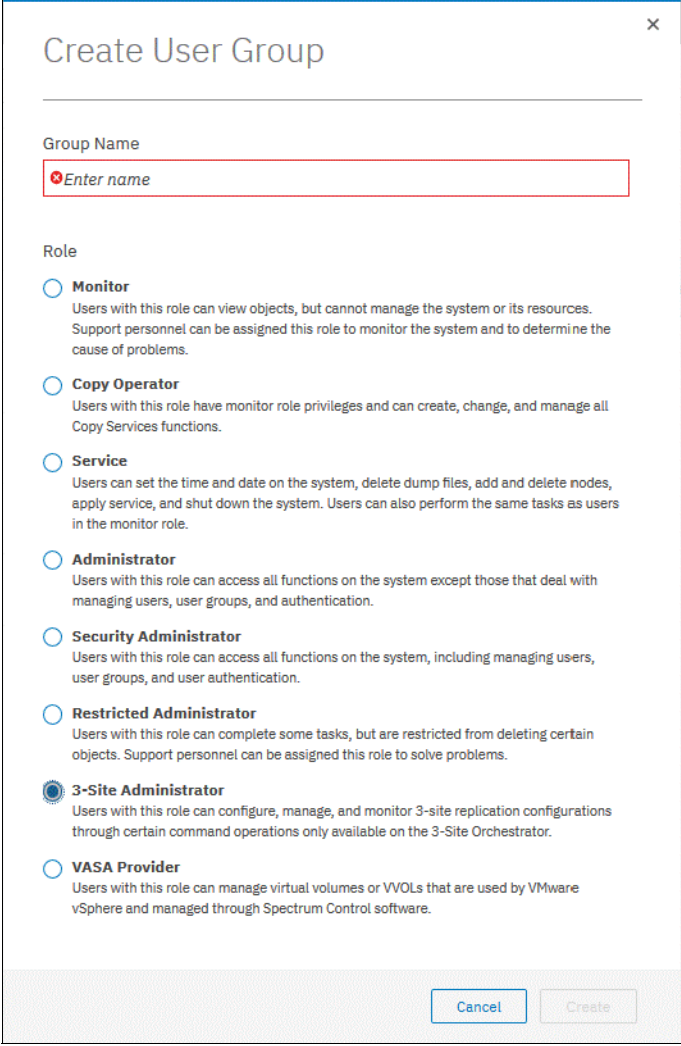


Figure 3-3 Create user group button

- ii. In a Create User Group dialog, specify the group's name and assign the 3-Site Administrator role, as shown in Figure 3-4 on page 29.



The image shows a 'Create User Group' dialog box with a title bar and a close button. It contains a 'Group Name' field with a red border and a red 'x' icon, with the text 'Enter name' inside. Below this is a 'Role' section with a list of roles, each with a radio button and a description. The roles are: Monitor, Copy Operator, Service, Administrator, Security Administrator, Restricted Administrator, 3-Site Administrator (which is selected), and VASA Provider. At the bottom right, there are 'Cancel' and 'Create' buttons.

Create User Group

Group Name

Enter name

Role

- ☐ **Monitor**
Users with this role can view objects, but cannot manage the system or its resources. Support personnel can be assigned this role to monitor the system and to determine the cause of problems.
- ☐ **Copy Operator**
Users with this role have monitor role privileges and can create, change, and manage all Copy Services functions.
- ☐ **Service**
Users can set the time and date on the system, delete dump files, add and delete nodes, apply service, and shut down the system. Users can also perform the same tasks as users in the monitor role.
- ☐ **Administrator**
Users with this role can access all functions on the system except those that deal with managing users, user groups, and authentication.
- ☐ **Security Administrator**
Users with this role can access all functions on the system, including managing users, user groups, and user authentication.
- ☐ **Restricted Administrator**
Users with this role can complete some tasks, but are restricted from deleting certain objects. Support personnel can be assigned this role to solve problems.
- ☒ **3-Site Administrator**
Users with this role can configure, manage, and monitor 3-site replication configurations through certain command operations only available on the 3-Site Orchestrator.
- ☐ **VASA Provider**
Users with this role can manage virtual volumes or VVOLs that are used by VMware vSphere and managed through Spectrum Control software.

Cancel Create

Figure 3-4 Create User Group dialog

- iii. After the group is created, click on the **Create User** button to see the dialog in Figure 3-5 on page 30. Set the user name that 3-Site Orchestrator will use, ensure that the User Group is set to one that was created with the previous step. Click **Browse** to select and upload the SSH public key that was created on a 3-Site Orchestrator system in c on page 25.

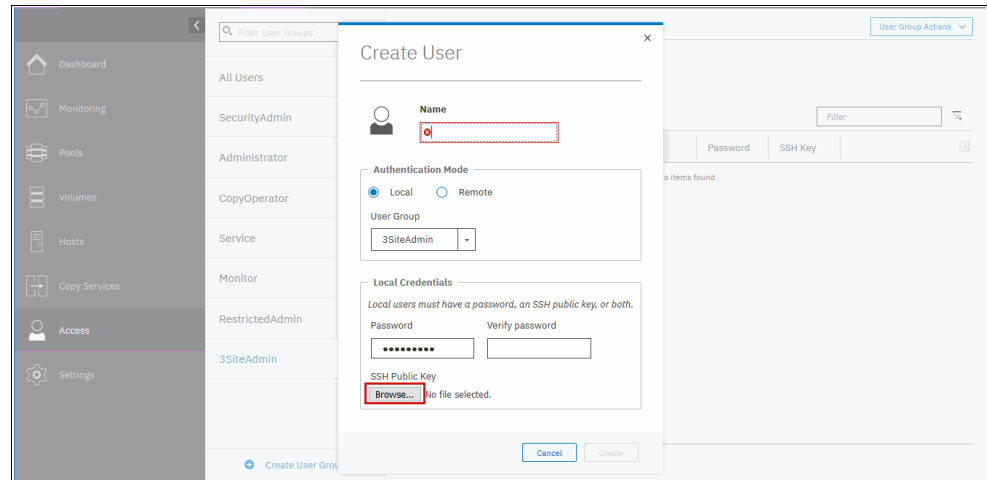


Figure 3-5 Create user dialog

- iv. Execute steps i on page 28, ii on page 28 and iii on the other two clusters. Note that the user name specified on all three clusters must match, and must use the same SSH public key.

3.3 Creating a 3-Site configuration

After the preparation steps are done, 3-Site replication configuration can be created. It is performed by running the 3-Site Orchestrator command **mk3siteconfig**. The command syntax is shown below, and the arguments are described in Table 3-1.

```
>> mk3siteconfig -- -ip1 site_ip1_list --+-----+----->
                                     '- -ip2 site_ip2_list --'
>---+ -username 3site_username ----- -keyfile ssh_keyfile_path ----->
>---+-----+-----+----- -port local_port_offset -----<
                                     '- -sitealias site_alias_list ---'
```

Table 3-1 mk3siteconfig command arguments

Argument	Description
ip1	(Required) Comma separated list of Spectrum Virtualize cluster IPv4 addresses for 3 sites. List should be in master, auxnear, auxfar order.
ip2	(Optional) Comma separated list of additional Spectrum Virtualize cluster IPv4 addresses for 3 sites. List should be in master, auxnear, auxfar order.
username	(Required) A 3-Site administrator user name defined on Spectrum Virtualize clusters on all 3 sites
keyfile	(Required) Path for private SSH key file for 3-Site administrator user

Argument	Description
sitealias	(Optional) A colon separated list of site alias for 3 sites. List should be in master, auxnear, auxfar order.
port	(Required) Starting address for 3 local ports to be used for connection with 3 sites over ssh tunnel. Orchestrator will use 3 consecutive ports starting with specified address.

Example 3-10 shows the **mk3siteconfig** command execution. If successful, it returns no output.

Example 3-10 mk3siteconfig command

```
[root@localhost ~]# mk3siteconfig -ip1 9.71.21.11,9.71.21.21,9.71.21.39 -username 3SiteOrch -keyfile /root/.ssh/3site -port 6000 -sitealias Master:AuxNear:AuxFar
```

When 3-Site configuration is created, you may check its status using the **ls3siteconfig** Orchestrator command. Example 3-11 shows the result.

Example 3-11 ls3siteconfig command

```
[root@localhost ~]# ls3siteconfig
master_name Master
master_ip1 9.71.21.11
master_ip1_status reachable
master_ip2
master_ip2_status not_configured
master_port 6000
auxnear_name AuxNear
auxnear_ip1 9.71.21.21
auxnear_ip1_status reachable
auxnear_ip2
auxnear_ip2_status not_configured
auxnear_port 6001
auxfar_name AuxFar
auxfar_ip1 9.71.21.39
auxfar_ip1_status reachable
auxfar_ip2
auxfar_ip2_status not_configured
auxfar_port 6002
username 3SiteOrch
keyfile /root/.ssh/3site
```

The 3-Site Orchestrator configuration is saved on an Orchestrator Linux host in an XML file at the location `'/opt/ibm/SpectrumVirtualize/rc3site/config.xml'`.

When a 3-Site configuration is created by 3-Site Orchestrator there will be no event indicating it in the Spectrum Virtualize cluster event log or audit log.

After the 3-Site configuration is created, you can start creating 3-Site replication.



Creating and managing a 3-Site Replication solution

In this chapter we describe the steps needed to start replicating data to the third site. It provides instructions for creating 2-Site replication pairs, and then converting them to 3-Site replication. Also, it contains instructions on converting 3-Site replication back to 2-Site replication.

This chapter requires the reader to have a knowledge of Spectrum Virtualize copy services. For more details on how they operate, refer to:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1*, SG24-8465
- ▶ *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1*, SG24-8466

This chapter contains the following topics:

- ▶ Replication configuration workflow
- ▶ Creating 2-Site replication
- ▶ Converting a 2-Site Consistency Group to 3-Site
- ▶ Adding a new relationship to an existing 3-Site Consistency Group
- ▶ Converting a 3-Site Consistency Group to 2-Site Metro Mirror
- ▶ Spectrum Virtualize GUI and CLI view of a 3-Site Replication

4.1 Replication configuration workflow

To start replicating data over three sites, the following are required:

- ▶ 3-Site configuration must be created.
- ▶ On Master and AuxNear sites, Metro Mirror replication relationships must be created, and Master and AuxNear volumes must be synchronized (so replication must be running in 2-Site mode).
- ▶ Relationships must be added into a Consistency Group (or groups). A standalone relationship can't be converted to 3-Site mode.
- ▶ By using 3-Site Orchestrator, the relationships of a Consistency Group must be converted, one by one, from 2-Site to 3-Site mode.

When all the relationships of the Consistency Group are in 3-Site mode, 3-Site Orchestrator starts data cycling automatically.

Instructions for creating the 3-Site configuration were provided in Chapter 3, “Configuring a 3-Site Replication solution” on page 23, instructions for the remaining actions are provided in the sections below.

4.2 Creating 2-Site replication

Before a replication can become 3-Site, it should be created as a “regular” 2-Site Metro Mirror replication. This section provides a brief overview of the process. For more details, refer to:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1*, SG24-8465
- ▶ *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1*, SG24-8466

If you already have a 2-Site Metro Mirror solution running, and you need to add a third site into the configuration, continue to 4.3, “Converting a 2-Site Consistency Group to 3-Site” on page 36.

To create a 2-Site Metro Mirror replication:

1. On the AuxNear system, create a volume that will contain a replica of the Master volume. The provisioned (available to host) capacity of a volume must be exactly same as the provisioned capacity of the Master volume. The volume type may be different, for example, if the Master volume is Fully Allocated, the AuxNear volume can be compressed.
2. On the Master system, create a Metro Mirror relationship:
 - a. In the GUI, navigate to **Copy Services** → **Remote Copy** and click **Create Consistency Group**, as shown in Figure 4-1. Specify the Consistency Group name and click **Next**.

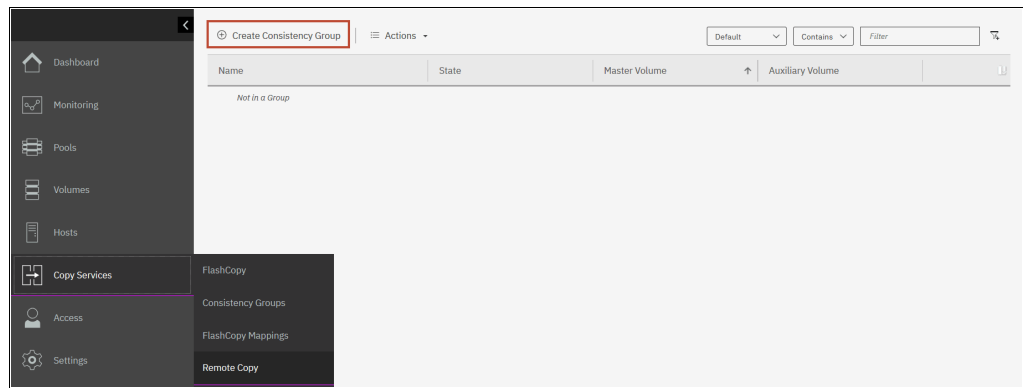


Figure 4-1 Creating a Consistency Group

- b. On the next page of the dialog, specify that the auxiliary volumes are located on another system, and select a system on the AuxNear site as shown in Figure 4-2. Click **Next**.

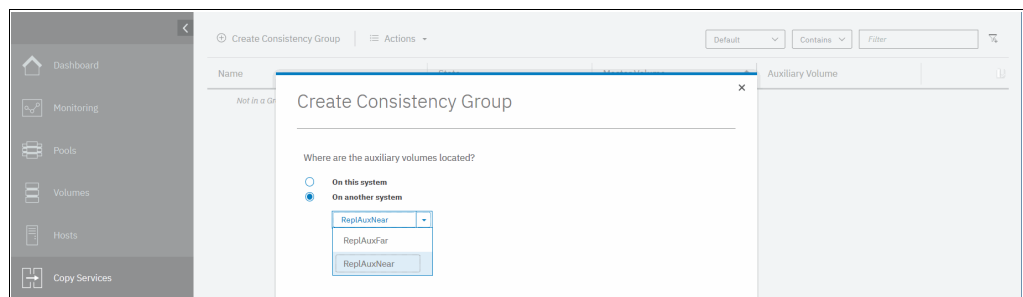


Figure 4-2 Auxiliary volumes location

- c. You will be asked if you want to add relationships to this Consistency Group. If you select **No**, an empty Consistency Group is created. Select **Yes** to start creating and adding relationships, and click **Next**.
- d. Select **Metro Mirror** as copy type. *Do not select* Add consistency protection. An example is shown in Figure 4-3. Click Next.

Note: The consistency protection feature for Metro Mirror is not supported with 3-Site replication. If you try to convert a relationship with consistency protection to 3-Site, the operation will fail with an error:

Operation failed with SVC error CMMVC9551E on system <Master IP>

You will need to delete the change volumes used by the consistency protection feature to allow you to convert the existing relationship to 3-Site.

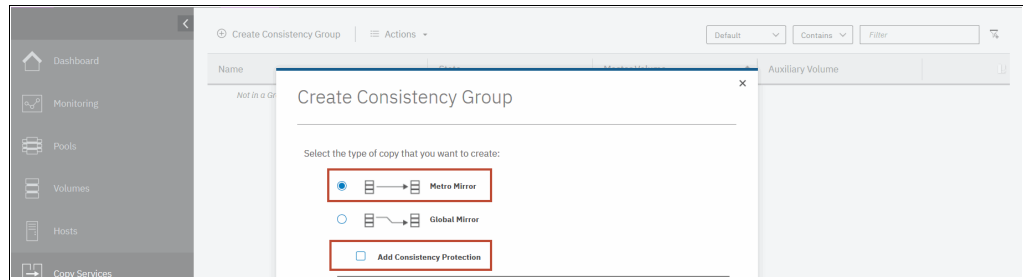


Figure 4-3 Selecting copy type

- e. If existing relationships need to be added to this Consistency Group, you can do it on the next screen of the dialog. Otherwise, click **Next** and select the Master and Auxiliary volumes that will be in a relationship, as shown in Figure 4-4.

Note: After you have selected a Master volume, only suitable candidates are displayed that are available for selection as an Auxiliary. If your desired auxiliary volume is not in the list, verify that it has exactly the same provisioned capacity as the Master volume.

You can create multiple relationships by clicking the **Add** button. When done, click **Next**.

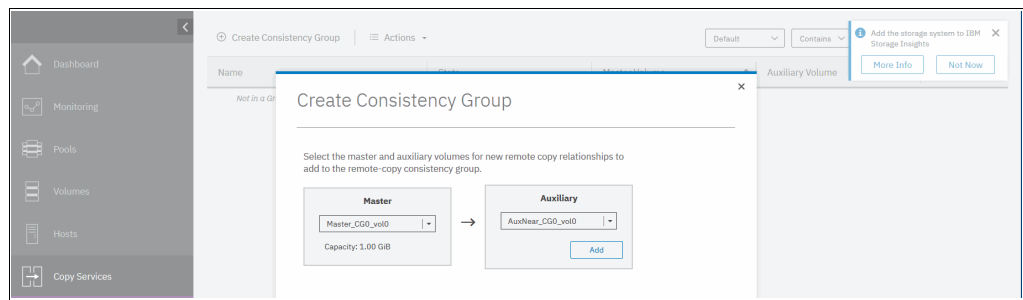


Figure 4-4 Selecting volumes for a relationship

- f. On the next screen, choose if synchronization is required. For example, if both Master and Auxiliary volumes are newly-created and formatted, no synchronization is needed. If your Master volume contains application data and Auxiliary is new, synchronization is required.
- g. At the next step, select **Yes** to start the Consistency Group.

If your relationships need synchronization, the Consistency Group will be in an *Inconsistent_Copying* state until synchronization is complete.

When the Consistency Group is in a *Consistent_Synchronized* state, it can be converted to 3-Site mode.

4.3 Converting a 2-Site Consistency Group to 3-Site

Before the third volume replicas can be added, determine which storage pool on the AuxFar system will be used to store them. Create new Standard or Data Reduction pools if required.

Replication relationships are migrated one by one. When the first relationship in a Consistency Group is converted, you need to specify which system (Master or AuxNear) will be the source for a replication to the third site, and a cycle period for periodic replication. All other converted relationships in the same Consistency Group will inherit those parameters. After all Consistency Group members are migrated to 3-Site mode, 3-Site replication starts automatically.

The 3-Site Orchestrator CLI is used to manage 3-Site replication.

Migration is done using the **convertrelationship** 3-Site Orchestrator CLI command. Its syntax is shown below, and the arguments are described in Table 4-1.

```
>> convertrelationship -- -type 3site -- -periodicsrc periodic_source_alias ->

>--- -cycleperiod cycleperiod ----- -pool auxfar_pool_id ----->

>---+-----+-----+-----+-----+-----+-----+-----+-----+-----+
      +- -iogrp auxfar_iogrp_id ---+      +- -volumename auxfar_volume_name -+

>---+-----+-----+-----+-----+-----+-----+-----+-----+-----+
      +- -thin -----+--+-----+--+-----+--+-----+--+-----+--+-----+--+
      '- -compressed --' '- -deduplicated --'
```

Table 4-1 3-Site Orchestrator convertrelationship arguments

Argument	Description
type	Intended type of target RC relationship post conversion. Can only be 3site , no other types are available (required parameter)
periodicsrc	Alias of a site or cluster name of a system that is the source for replication to the third site. Site aliases are defined when the 3-Site configuration is created, ls3siteconfig CLI can be used to list them. This parameter is required for the first relationship being converted within a 2-Site Consistency Group
cycleperiod	Cycle period in seconds for periodic replication. Minimum supported value is 300 (5 minutes). This parameter is required for the first relationship being converted within a 2-Site Consistency Group
pool	Storage pool id where the replica volume is created on the AuxFar site (required parameter)
iogrp	Caching IO group id for the new volume on the AuxFar site. If not specified, IO group 0 is assigned (optional parameter)
volumename	Specifies the name for the new volume on the AuxFar site. If not set, <i>volumeXX</i> name is given (optional parameter)

Argument	Description
thin compressed deduplicated	Specifies the volume type for the AuxFar site. It can be thin, compressed, thin deduplicated, or compressed deduplicated. If none of the parameters are set, the volume is created as fully allocated (optional parameter)
relationship_name	Relationship name of the 2-Site replication relationship that is being converted (required parameter)

For example, to convert a 2-Site Consistency Group, the following commands may be used:

- From a Master system CLI, we can find out that a Consistency Group to be converted CG0_3site, id 0, consists of four Metro Mirror relationships (rcrel0-rcrel3). It is in a consistent_synchronized state as Example 4-1 shows. The replication primary system is “master”, so application data is currently accessible from the cluster named ReplMaster.

Example 4-1 2-Site replication relationships

```
IBM_Spectrum_Virtualize:ReplMaster:superuser>lsrconsistgrp 0
id 0
name CG0_3site
master_cluster_name ReplMaster
aux_cluster_name ReplAuxNear
primary master
state consistent_synchronized
relationship_count 4
copy_type metro
cycle_period_seconds 300
RC_rel_id 0
RC_rel_name rcrel0
RC_rel_id 1
RC_rel_name rcrel1
RC_rel_id 2
RC_rel_name rcrel2
RC_rel_id 3
RC_rel_name rcrel3
```

- From the 3-Site Orchestrator CLI, we can start converting a Consistency Group to 3-Site mode by converting the first relationship (rcrel0), as demonstrated in Example 4-2.

On the AuxFar site, 3-Site Orchestrator will create a thin-provisioned volume in Pool 0.

The **ls3sitercrelationship** command shows that the 3-Site relationship has a status of COMPLETE, which means that all the necessary components on all three clusters are created and ready to operate.

The **ls3sitercconsistgrp** command shows that the Consistency Group is in a partial state. It indicates that the Consistency Group, or any of the relationships in it, are currently incomplete. Also the `periodic_copy_state` is none, as the 3-Site Consistency Group will not start copying data to the third site before all 2-Site relationships are converted.

Example 4-2 Converting the first relationship

```
[root@localhost ~]# converttrcrelationship -type 3site -cycleperiod 300
-periodicsrc Master -pool 0 -thin rcrel0
3-site RC relationship created with volume id [0].
[root@localhost ~]# ls3sitercrelationship
name consistgrp master_vdisk_id auxnear_vdisk_id auxfar_vdisk_id
sync_copy_progress periodic_copy_progress status
rcrel0 CG0_3site 0 0 0
COMPLETE
```

```
[root@localhost ~]# ls3sitercconsistgrp
name      state      primary periodic_copy_source sync_copy_state
periodic_copy_state freeze_time cycle_period status
CG0_3site partial Master Master          consistent_synchronized None
None      300
```

- From the 3-Site Orchestrator CLI, the remaining 2-Site relationships are converted with the same **converttrcrelationship** command. The remaining relationships in the source Consistency Group can be converted without specifying the **-cycleperiod** and **-periodicsrc** parameters. Example 4-3 shows the conversion commands and their result.

When all four relationships of the 2-Site Consistency Group are migrated and shown by 3-Site Orchestrator in COMPLETE state, the 3-Site Consistency Group changes its status to 3site_inconsistent, and the first data replication cycle starts. It may take a significant amount of time for initial synchronization with the third site.

Example 4-3 Completing Consistency Group conversion to 3-Site

```
[root@localhost ~]# converttrcrelationship -type 3site -pool 0 -thin rcrel1
3-site RC relationship created with volume id [3].
[root@localhost ~]# converttrcrelationship -type 3site -pool 0 -thin rcrel2
3-site RC relationship created with volume id [6].
[root@localhost ~]# converttrcrelationship -type 3site -pool 0 -thin rcrel3
3-site RC relationship created with volume id [9].
[root@localhost ~]# ls3sitercconsistgrp
name      consistgrp master_vdisk_id auxnear_vdisk_id auxfar_vdisk_id
sync_copy_progress periodic_copy_progress status
rcrel0 CG0_3site 0 0 0
0 COMPLETE
rcrel1 CG0_3site 1 1 3
0 COMPLETE
rcrel2 CG0_3site 2 2 6
0 COMPLETE
rcrel3 CG0_3site 3 3 9
0 COMPLETE

[root@localhost ~]# ls3sitercconsistgrp
name      state      primary periodic_copy_source sync_copy_state
periodic_copy_state freeze_time cycle_period status
CG0_3site 3site_inconsistent Master Master
consistent_synchronized copying 300 online
```

- After the first cycle completes successfully, the 3-Site Consistency Group becomes 3site_consistent, as Example 4-4 shows. Periodic_copy_state will remain stopped until the next replication cycle.

Example 4-4 Running 3-Site Consistency Group

```
[root@localhost ~]# ls3sitercconsistgrp
name      state      primary periodic_copy_source sync_copy_state
periodic_copy_state freeze_time cycle_period status
CG0_3site 3site_consistent Master Master          consistent_synchronized
stopped 2020/03/31/11/02/28 300 online
```

4.4 Adding a new relationship to an existing 3-Site Consistency Group

You can add a new standalone 2-Site replication relationship to an existing 3-Site Consistency Group. This procedure has the following prerequisites:

- ▶ The 3-Site Consistency Group must be in a `3site_consistent` state;
- ▶ The standalone relationship must be created between the newly created empty volumes with no application data on them;
- ▶ The state of the standalone relationship that needs to be added must be `consistent_synchronized`;
- ▶ All the attributes of a 2-Site relationship that needs to be added must match with the target 3-Site Consistency Group;
- ▶ There must be no IO operations running on the 2-Site relationship that is being added.

The command **addnewrelationship** is used for this task. Its syntax is shown below, and the parameters are described in Table 4-2.

```
>> addnewrcrelationship -- -consistgrpname 3siteCGname -- -pool auxfar_pool -->
```

```
>---+-----+-----+-----+-----+-----+-----+-----+-----+-->  
      +- iogrp auxfar_iogrp_id ---+          +- -volumename auxfar_volume_name -+  
  
>---+-----+-----+-----+-----+-----+-----+-----+-----+-->  
      +- thin -----+--+-----+-----+-----+-'  
      '- compressed --'    '- deduplicated --'
```

Table 4-2 CLI addnewrcrelationship arguments

Argument	Description
consistgrpname	The name of the existing 3-Site Consistency Group to which a relationship is added (required parameter)
pool	The Storage pool id where the replica volume is created on the AuxFar site (required parameter)
iogrp	The caching IO group id for the new volume on the AuxFar site. If not specified, IO group 0 is assigned (optional parameter)
volumename	Specifies the name for the new volume on the AuxFar site. If not set, <i>volumeXX</i> name is given (optional parameter)
thin compressed deduplicated	Specifies the volume type for the AuxFar site. It can be thin, compressed, thin deduplicated, or compressed deduplicated. If none of the parameters are set, the volume is created as fully allocated (optional parameter)
relationship_name	The relationship name of the 2-Site replication relationship that is added (required parameter)

Review Example 4-5 to see how a single relationship is added to a Consistency Group.

- ▶ Create new volumes on the Master and AuxNear systems with matching provisioned capacities.
- ▶ Create a 2-Site Metro Mirror relationship between those volumes. Note that the relationship primary cluster must match to the primary cluster of the target 3-Site Consistency Group.
- ▶ Make sure that the new standalone relationship is in a `consistent_synchronized` state, as shown in Example 4-5.

Example 4-5 Listing relationships

```
IBM_Spectrum_Virtualize:ReplMaster:superuser>lsrcrelationship -delim " "
id name master_cluster_name master_vdisk_id master_vdisk_name aux_cluster_name
aux_vdisk_id aux_vdisk_name primary consistency_group_id consistency_group_name
state bg_copy_priority progress copy_type
0 rcrel0 ReplMaster 0 Master_CG0_vol0 ReplAuxNear 0 AuxNear_CG0_vol0 master 0
CG0_3site consistent_synchronized 50 metro
[...]
4 rcrel4 ReplMaster 4 Master_vol4 ReplAuxNear 4 AuxNear_vol4 master
consistent_synchronized 50 metro
```

- ▶ In the 3-Site Orchestrator CLI, use the **addnewrcrelationship** command to add the relationship to an existing 3-Site Consistency Group, as shown in Example 4-6.

Example 4-6 Adding new relationship

```
[root@localhost ~]# addnewrcrelationship -consistgrpname CG0_3site -pool 0
-thin rcrel4
3-site RC relationship created with volume id [12].
```

4.5 Converting a 3-Site Consistency Group to 2-Site Metro Mirror

If needed, the 3-Site Consistency Group may be converted back to 2-Site replication running between the Master and AuxNear sites. All relationships within the 3-Site Consistency Group will be converted. Data cycling must be stopped prior to conversion.

After the 3-Site Consistency Group is converted, all volumes on the AuxFar site will remain, and become accessible. All internal objects that were used for 3-Site replication are deleted.

The command **converttrconsistgrp** is used for this task. Its syntax is shown below, and its arguments are described in Table 4-3.

```
>> converttrconsistgrp -- -type -2site- ----- 3-site_consistency_group_name
```

Table 4-3 CLI **converttrconsistgrp** arguments

Argument	Description
type	The intended type of the target RC relationship post conversion. It can only be <i>2site</i> , no other types are available (required parameter)

Argument	Description
group name	The name of a 3-Site Consistency Group that is converted to 2-Site (required parameter)

For conversion to 2-Site to be possible, the state of the 3-Site Consistency Group must be stopped or partial. To stop it, the 3-Site Orchestrator **stop3sitercconsistgrp** command must be issued. The only required argument for this command is the name of the Consistency Group that needs to be stopped.

Review Example 4-7 of a 3-Site to 2-Site Consistency Group conversion.

- Initially, the 3-Site Consistency Group is working, as shown in Example 4-7, so the **converttrcconsistgrp** command returns an error.

Example 4-7 State of a 3-Site Consistency Group

```
[root@localhost ~]# ls3sitercconsistgrp
name      state      primary periodic_copy_source sync_copy_state
periodic_copy_state freeze_time      cycle_period status
CG0_3site 3site_consistent Master Master      consistent_synchronized
stopped      2020/03/31/13/53/32 300      online
[root@localhost ~]# converttrcconsistgrp -type 2site CG0_3site
CMMVC9518E State or type of specified 3-site consistency group is not valid.
```

- After we issue **stop3sitercconsistgrp**, the group changes to a stopped state and can be converted, as shown in Example 4-8. CG0_3site was the only 3-Site group configured, so the **ls3sitercconsistgrp** returns no output.

Example 4-8 Converting Consistency Group to 2-Site

```
[root@localhost ~]# stop3sitercconsistgrp CG0_3site
[root@localhost ~]# ls3sitercconsistgrp
name      state      primary periodic_copy_source sync_copy_state
periodic_copy_state freeze_time      cycle_period status
CG0_3site stopped Master Master      consistent_synchronized idling
2020/03/31/13/58/32 300      online

[root@localhost ~]# converttrcconsistgrp -type 2site CG0_3site

[root@localhost ~]# ls3sitercconsistgrp
[root@localhost ~]#
```

4.6 Spectrum Virtualize GUI and CLI view of a 3-Site Replication

All objects used for 3-Site replication must only be managed by 3-Site Orchestrator. However, by using the Spectrum Virtualize CLI and GUI, the system administrator can detect 3-Site configuration activity.

Important: Do not delete or alter any Consistency Groups or relationships that are used in a 3-Site solution with the GUI or CLI of a Spectrum Virtualize system. All administrator actions must be performed using the 3-Site Orchestrator CLI.

If 3-Site Consistency Groups are configured, the GUI **Remote Copy** panel will show a warning, stating that 3-Site Orchestrator must be used for management. An example from the 3-Site replication Master system is shown in Figure 4-5.

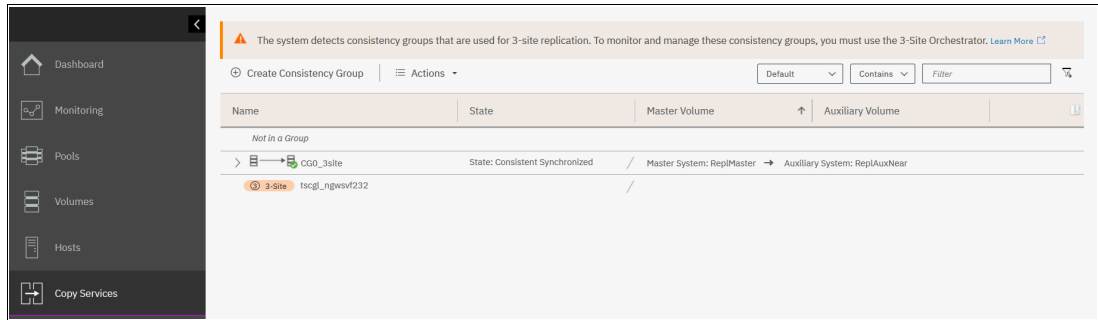


Figure 4-5 Remote Copy panel detects 3-Site configuration objects

Objects used for 3-Site replication operations, such as AccessPoint volumes and their FlashCopy mappings, are not shown in the GUI. However, using the CLI you can list them and see their properties. Example 4-9 shows `lsvdisk` output from an AuxFar site. You can see volume id 1, volume5, that will also be visible in the GUI, and used to store the data, and two AccessPoint vdisks. AccessPoint vdisks have a corresponding `owner_type` `accesspoint`.

Example 4-9 Listing volumes on AuxFar site

```
IBM_Spectrum_Virtualize:ReplAuxFar:superuser>lsvdisk
id name      IO_group_id IO_group_name status  mdisk_grp_id mdisk_grp_name capacity
1  volume5  0          io_grp0    online  0            AuxFar_Poo10  1.00GB
2  vdisk0   0          io_grp0    online  0            AuxFar_Poo10  1.00GB
4  vdisk1   0          io_grp0    offline 0            AuxFar_Poo10  1.00GB
IBM_Spectrum_Virtualize:ReplAuxFar:superuser>lsvdisk vdisk0
id 2
name vdisk0
[...]
RC_id 2
RC_name tsrell_cxhwpt84
fc_map_count 2
[...]
owner_type accesspoint
```

Also, using the Spectrum Virtualize CLI, you can see the Consistency Groups and remote copy relationships that the 3-Site solution uses to maintain the third copy. Example 4-10 shows the Consistency Group and relationship listings. Note the naming of the objects created by 3-Site Orchestrator: for example, `tscgl_XXXXX` means that it is a **three site consistency group - left**.

Example 4-10 Listing 3-Site objects with CLI

```
IBM_Spectrum_Virtualize:ReplAuxFar:superuser>lsrcconsistgrp
id name          master_cluster_id master_cluster_name aux_cluster_id
0  tscgl_ngwsvf232 0000001C61200002 ReplMaster         0000001C65200018
1  tscgr_ngwsvf232 0000001C6EE00020 ReplAuxNear        0000001C65200018
IBM_Spectrum_Virtualize:ReplAuxFar:superuser>lsrcrelationship
id name          master_cluster_id master_cluster_name master_vdisk_id
master_vdisk_name
2  tsrell_cxhwpt84 0000001C61200002 ReplMaster         10                vdisk0
4  tsrelr_cxhwpt84 0000001C6EE00020 ReplAuxNear        5                 vdisk0
```

7	tsrell_ueoatk32	0000001C61200002	ReplMaster	11	vdisk1
8	tsre1r_ueoatk32	0000001C6EE00020	ReplAuxNear	6	vdisk1

With the **Access** → **Audit Log** GUI panel, you can see the commands that the 3-Site Orchestrator issues to the system. Figure 4-6 shows the commands that 3-Site Orchestrator runs on the AuxFar site to convert one Consistency Group containing five relationships.

Date and Time	User Name	Command	Object ID
3/31/2020 7:08:55 PM	3SiteOrch	svctask chrconsistgrpaccesspoint -channel left CG0_3site	
3/31/2020 7:08:53 PM	3SiteOrch	svctask addvdiskaccesspoint -channel right 17	19
3/31/2020 7:08:53 PM	3SiteOrch	svctask addvdiskaccesspoint -channel left 17	18
3/31/2020 7:08:53 PM	3SiteOrch	svctask mkvolume -size 1073741824 -unit b -pool 0 -thin	17
3/31/2020 7:08:46 PM	3SiteOrch	svctask addvdiskaccesspoint -channel right 14	16
3/31/2020 7:08:45 PM	3SiteOrch	svctask mkvolume -size 1073741824 -unit b -pool 0 -thin	14
3/31/2020 7:08:45 PM	3SiteOrch	svctask addvdiskaccesspoint -channel left 14	15
3/31/2020 7:08:38 PM	3SiteOrch	svctask addvdiskaccesspoint -channel left 10	11
3/31/2020 7:08:38 PM	3SiteOrch	svctask addvdiskaccesspoint -channel right 10	13
3/31/2020 7:08:38 PM	3SiteOrch	svctask mkvolume -size 1073741824 -unit b -pool 0 -thin	10
3/31/2020 7:08:31 PM	3SiteOrch	svctask addvdiskaccesspoint -channel right 5	8
3/31/2020 7:08:30 PM	3SiteOrch	svctask mkvolume -size 1073741824 -unit b -pool 0 -thin	5
3/31/2020 7:08:30 PM	3SiteOrch	svctask addvdiskaccesspoint -channel left 5	7
3/31/2020 7:08:13 PM	3SiteOrch	svctask mkvolume -size 1073741824 -unit b -pool 0 -thin	1
3/31/2020 7:08:13 PM	3SiteOrch	svctask addvdiskaccesspoint -channel left 1	2
3/31/2020 7:08:13 PM	3SiteOrch	svctask addvdiskaccesspoint -channel right 1	4
3/31/2020 7:08:12 PM	3SiteOrch	svctask mkrcconsistgrpaccesspoint -name CG0_3site -leftgrp tsc...	0

Figure 4-6 Audit log showing Orchestrator activity



3-Site Replication Monitoring and Maintenance

This chapter describes monitoring 3-Site Replication including the following:

- ▶ Monitoring using the 3-Site Orchestrator
- ▶ 3-Site Replication log files
- ▶ Orchestrator events shown at the AuxFar cluster
- ▶ Maintenance tasks
- ▶ Upgrading the storage-system software in a 3-Site configuration

5.1 Orchestrator monitoring commands

3-Site Orchestrator provides three commands to check the configuration of the Orchestrator and to check the status of the 3-Site relationships and Consistency Groups (CGs). These commands are:

► **ls3siteconfig**

The command **ls3siteconfig** lists the Orchestrator configuration.

► **ls3siterconsistgrp**

The command **ls3siterconsistgrp** lists the detailed status for all 3-Site remote copy Consistency Groups present in a 3-Site configuration.

► **ls3sitercrelationship**

The command **ls3sitercrelationship** lists the detailed status of all 3-Site remote copy relationships present in a 3-Site configuration.

The 3-Site Orchestrator service can run only one command at a time as shown in Example 5-1.

Example 5-1 Orchestrator command error

```
[root@localhost ~]# ls3sitercrelationship
CMMVC9481E 3-Site orchestrator is busy processing another command.
```

If the error *CMMVC9481E* is shown, repeat the command until it succeeds.

5.1.1 List and check the Orchestrator configuration

The command **ls3siteconfig** lists the Orchestrator configuration as shown in Example 5-2.

Example 5-2 3-Site Orchestrator configuration

```
[root@localhost ~]# ls3siteconfig
master_name Master
master_ip1 10.10.122.238
master_ip1_status reachable
master_ip2
master_ip2_status not_configured
master_port 6000
auxnear_name AuxNear
auxnear_ip1 10.10.114.16
auxnear_ip1_status reachable
auxnear_ip2
auxnear_ip2_status not_configured
auxnear_port 6001
auxfar_name AuxFar
auxfar_ip1 10.10.113.247
auxfar_ip1_status reachable
auxfar_ip2
auxfar_ip2_status not_configured
auxfar_port 6002
username 3SiteUser
keyfile /root/.ssh/id_rsa
```

Every cluster has the first IP address configured and in this example the second IP address is not configured. The possible values for the IP connection status are:

- ▶ `reachable`
One IP address of a cluster must have the status `reachable`. Otherwise the Orchestrator can not perform any 3-Site activity, like starting the mirroring to the third site.
- ▶ `unreachable`
The IP connection to the cluster should be checked for errors.
- ▶ `not_configured`
One cluster IP-address has to be configured. The other cluster IP address can remain unconfigured.
- ▶ `authentication_failed`
The 3-Site username and the 3-Site keyfile listed in the `ls3siteconfig` command output should be checked.

Orchestrator services

The following three 3-Site Orchestrator services must be running:

- ▶ **Task Daemon**
The service `/usr/bin/tstaskd` controls the replication of the configured 3-Site Orchestrator Consistency Groups. The log file of the 3-Site task daemon is:
`/opt/ibm/SpectrumVirtualize/rc3site/logs/tstaskd.log`
- ▶ **Event Daemon**
The service `/usr/bin/eventd` communicates with all three sites and monitors various important attributes of the sites and configured Consistency Groups. Port number 7003 is reserved for the eventd service. The events are sent to the AuxFar cluster, and are available using the GUI or CLI as shown in 5.3, “Event logging at the AuxFar site” on page 54.
- ▶ **Miscellaneous Daemon**
The service `/usr/bin/tsmiscservicesd` maintains an audit log of all 3-Site Orchestrator commands that are issued. The log file of the 3-Site miscellaneous daemon is:
`/opt/ibm/SpectrumVirtualize/rc3site/logs/tsmiscservicesd.log`

The status of the three services can be checked using the `ps` and `systemctl` commands as shown in Example 5-3.

Example 5-3 Service status check

```
[root@localhost ~]# ps -ef | grep -v grep | egrep -e '(eventd|tstaskd|tsmiscservicesd)'
root      2011      1  0 Mar24 ?        00:00:07 /usr/bin/eventd 7003
root      2017      1  0 Mar24 ?        00:12:32 /usr/bin/tsmiscservicesd start
root      2028      1  0 Mar24 ?        00:14:19 /usr/bin/tstaskd start
```

```
[root@localhost ~]# systemctl | grep -v grep | egrep -e '(eventd|tstaskd|tsmiscservicesd)' | grep rc3site
rc3site.eventd.service          loaded active running  rc3site eventd service.
rc3site.tsmiscservicesd.service loaded active running  rc3site tsmiscservicesd service.
rc3site.tstaskd.service         loaded active running  rc3site tstaskd service.
```

The three listed processes must always be active, and running, to ensure a correctly working Orchestrator.

5.1.2 Monitoring the 3-Site Consistency Groups

The command **ls3sitercconsistgrp** lists the detailed status of all 3-Site remote copy Consistency Groups present in a 3-Site configuration.

Example 5-4 shows the output of the **ls3sitercconsistgrp** command after data has been transferred to the AuxFar site.

Example 5-4 3-Site Consistency Group information

```
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time cycle_period status
CG0 3site_consistent Master Master consistent_synchronized stopped 2020/03/24/19/14/13 300 online
```

The attributes of the **ls3sitercconsistgrp** command are:

► **name**

The name of a 3-Site remote copy Consistency Group.

► **state**

This state indicates the state of 3-Site data cycling. The possible values are:

– **3site_consistent (3SC)**

Indicates that a consistent copy of data is available on the AuxFar site. The AuxFar site will be within a maximum of two cycle periods of the application and meet the defined Recovery Point Objective (RPO).

– **3site_inconsistent (3SI)**

Indicates that data on the AuxFar site volume is inconsistent with data on the near sites. This may be the result of errors, or the first data cycling is still ongoing and has not yet completed.

– **2site_periodic (2SP)**

Indicates that the non-primary near site is not part of the 3-Site configuration. The primary near site can replicate data to the AuxFar site, and a consistent copy of data is available on the AuxFar site. The AuxFar site will be within a maximum of two cycle periods of the application.

– **partial**

The 3-Site replication for at least one relationship of the CG is not fully configured. This could be the case if an error occurred during the 3-Site configuration, like using an inappropriate type for third site volume, or the near sites have not been synchronized, or other errors.

After the errors are fixed, the 3-Site CG configuration command can be repeated and will complete the configuration.

– **stopped**

Indicates that 3-Site data cycling is stopped, due to a link failure, disaster, or by issuing the **stop3sitercconsistgrp** command. The command **stop3sitercconsistgrp** will stop the periodic cycling of a CG and this status will be shown. While 3-Site data cycling is stopped, the recovery point objective (RPO) of the auxiliary-far site copy of data is extended by the duration for which 3-Site data cycling is stopped.

► **primary**

Specifies the primary site, either Master or AuxNear, for the 3-Site remote copy Consistency Group. This value is the site's alias cluster name.

► **periodic_copy_source**

Specifies the source site for periodic remote copy site, either Master or AuxNear. The value is the site's alias cluster name.

► `sync_copy_state`

Specifies the state of the near site synchronous copy. The possible values are:

- `consistent_synchronized`
- `idling_disconnected`
- `idling`
- `consistent_copying`
- `consistent_stopped`

► `periodic_copy_state`

Specifies the state of the remote site periodic copy. The possible values are:

- `copying`
This is normal state during a periodic data cycling. The data from periodic source AccessPoint is transferred to the third site.
- `copied`
This is normal state after a periodic data cycling has transferred all data.
- `stopped`
This is normal state after a periodic data cycling has ended.
- `idling`
This is normal state between copied and stopped. For example, to set the correct `freeze_time` value.

► `freeze_time`

Specifies the date of the consistent copy at the AuxFar site. The time stamp is the event time at the AuxFar site.

► `cycle_period`

Specifies the 3-Site replication cycle period in seconds.

► `status`

Specifies the 3-Site data cycling status for the 3-Site CG. The status online indicates a healthy 3-Site cycling. Every other status indicates an error which has to be fixed. The possible status values are:

- `online`
3-Site data cycling is working in `3site_consistent` mode.
- `sync_link_offline`
The near site link is disconnected. To ensure a consistent copy at the AuxFar site make sure that the primary site is the periodic source.
- `sync_link_stopped`
The near site link is connected but it is in a stopped state. To assure a consistent copy at the AuxFar site make sure that the primary site is the periodic source.
- `active_link_offline`
The active link is down, and 3-Site data cycling is in a stopped state. If the link error can not be fixed immediately, then move the periodic source site to continue in degraded mode.

- `inactive_link_offline`

The idle link is down.

- `primary_offline`

This value indicates that the primary site for this Consistency Group is down or is isolated from the partnership (there is no available network and the site is unable to be reached from the remaining two sites). No data will be replicated to the AuxFar site until this error is fixed.

- `periodic_src_offline`

This value indicates that the periodic site for this Consistency Group is down or is isolated from the partnership (there is no available network and the site is unable to be reached from the remaining two sites). If the primary site is online then move the periodic source site to the primary site and continue in degraded mode.

- `primary_and_periodic_src_offline`

This value indicates that the primary and periodic source sites for this Consistency Group are down or isolated from the partnership. The 3-Site Consistency Group is in the stopped state. No data will be replicated to the AuxFar site until this error is fixed.

- `partner_offline`

This value indicates that the near partner site, which is neither a primary nor periodic source, is offline. Data cycling is in `2site_periodic` mode.

- `auxfar_offline`

This value indicates that the AuxFar site is offline. The data copies at the AuxFar site are not accessible.

- `primary_storage_offline`

The 3-Site relationship is either in the stopped or `2site_periodic` state. The primary site storage is offline. To ensure a consistent copy at the AuxFar site, make sure that the secondary site is the periodic source.

- `periodic_src_storage_offline`

The 3-Site relationship is either in the stopped or `2site_periodic` state. The periodic site storage is offline. If the periodic source is not the primary site, then make sure that the primary site is the periodic source and replicates data to the AuxFar site.

- `partner_storage_offline`

This value indicates that at the near partner site, which is neither a primary nor a periodic source, storage is offline. Data cycling is in `2site_periodic` mode.

- `auxfar_storage_offline`

This value indicates that at the AuxFar site the storage is offline. Data cycling is stopped.

- `site_unreachable`

This value indicates one or more sites lost connection with 3-Site Orchestrator. Data cycling resumes automatically after the error is fixed.

- `maintenance_exclusion_mode`

This value indicates that a site is excluded for maintenance and its state is `2site_periodic` or stopped.

If the primary site is down, then a disaster recovery at the AuxNear site has to be started. A possible scenario is to move the workload to the non primary site and to use the new primary site as the periodic source in a degraded state.

Note: Depending on the event, or the error that has occurred at one of the three sites, the corresponding information is not immediately propagated to the Orchestrator. The **ls3sitercconsistgrp** command will show the state change some minutes after it is seen at one of the clusters.

Example 5-5 shows the periodic copy states during a cycle period. Some command output values are omitted for better readability.

Example 5-5 Periodic copy states during a cycle period.

[root@localhost ~]# ## time-stamp and ls3sitercconsistgrp output

Time:	name	primary	sync_copy_state	periodic_copy_state	freeze_time
15:15:57:	CG0	Master	consistent_synchronized	copying	2020/03/30/14/10/46
15:18:20:	CG0	Master	consistent_synchronized	copied	2020/03/30/14/10/46
15:18:23:	CG0	Master	consistent_synchronized	idling	2020/03/30/14/10/46
15:18:24:	CG0	Master	consistent_synchronized	idling	2020/03/30/14/15/51
15:18:26:	CG0	Master	consistent_synchronized	stopped	2020/03/30/14/15/51

Example 5-6 shows the state of the 3-Site CG (attribute *state*), the state of the near site synchronous copy (attribute *sync_copy_state*), and the 3-Site data cycling status (attribute *status*) during the failure of the link between the near sites in a Cascade topology.

Some command output values are omitted for better readability.

Example 5-6 3-Site data cycling status for the 3-Site CG CG0 while the link between Master and AuxNear goes offline.

[root@localhost ~]# ## time-stamp and ls3sitercconsistgrp output

	name	state	periodic_copy_source	sync_copy_state	periodic_copy_state	freeze_time	status
16:31:00:	CG0	3site_consistent	AuxNear	consistent_synchronized	stopped	2020/03/30/15/28/31	online
16:32:07:	CG0	3site_consistent	AuxNear	idling_disconnected	stopped	2020/03/30/15/28/31	online
16:32:31:	CG0	3site_consistent	AuxNear	idling_disconnected	stopped	2020/03/30/15/28/31	sync_link_offline
16:32:31:	CG0	stopped	AuxNear	idling_disconnected	stopped	2020/03/30/15/28/31	sync_link_offline

After the link failure, the periodic source is changed to primary (Star topology) using the **ch3sitercconsistgrp** command. This change restores the cycling to the third site as shown in Example 5-7. The new state is 2site_periodic, and therefore the AuxFar site will be within a maximum of two cycle periods of the application.

Some command output values are omitted for better readability.

Example 5-7 3-site data cycling status for the 3-Site CG CG0 while the Near Sites links goes offline.

[root@localhost ~]# **ch3sitercconsistgrp** -periodicsrc Master CG0

[root@localhost ~]# ## time-stamp and ls3sitercconsistgrp output

	name	state	periodic_copy_source	sync_copy_state	periodic_copy_state	freeze_time	status
16:57:30:	CG0	stopped	AuxNear	idling_disconnected	stopped	2020/03/30/15/55/01	sync_link_offline
16:57:33:	CG0	stopped	AuxNear	idling_disconnected	idling	2020/03/30/15/55/01	sync_link_offline
16:57:34:	CG0	stopped	Master	idling_disconnected	idling	2020/03/30/15/55/01	sync_link_offline
16:57:36:	CG0	2site_periodic	Master	idling_disconnected	idling	2020/03/30/15/55/01	sync_link_offline
16:57:42:	CG0	2site_periodic	Master	idling_disconnected	stopped	2020/03/30/15/55/01	sync_link_offline
16:57:45:	CG0	2site_periodic	Master	idling_disconnected	copied	2020/03/30/15/55/01	sync_link_offline
16:57:46:	CG0	2site_periodic	Master	idling_disconnected	idling	2020/03/30/15/55/01	sync_link_offline
16:57:48:	CG0	2site_periodic	Master	idling_disconnected	stopped	2020/03/30/15/57/41	sync_link_offline

5.1.3 Monitoring the 3-Site Remote Copy relationships

The command **ls3sitercrelationship** lists the detailed status of all 3-Site remote copy relationships present in a 3-Site configuration.

Example 5-8 shows the output of the **ls3sitercrelationship** command after data has been transferred to the AuxFar site.

Example 5-8 3-Site Consistency Group information

```
[root@localhost ~]# ls3sitercrelationship
```

name	consistgrp	master_vdisk_id	auxnear_vdisk_id	auxfar_vdisk_id	sync_copy_progress	periodic_copy_progress	status
rcre10	CG0	12	12	0		100	COMPLETE
rcre11	CG0	13	13	3		100	COMPLETE

The attributes of the **ls3sitercconsistgrp** command are:

- ▶ **name**
Specifies the name of the 3-Site remote copy relationship.
- ▶ **consistgrp**
Specifies the name of the parent 3-Site remote copy Consistency Group.
- ▶ **master_vdisk_id**
Specifies the master volume ID in the 3-Site remote copy relationship.
- ▶ **auxnear_vdisk_id**
Specifies the auxiliary-near volume ID in the 3-Site remote copy relationship.
- ▶ **auxfar_vdisk_id**
Specifies the auxiliary-far volume ID in the 3-Site remote copy relationship.
- ▶ **sync_copy_progress**
Specifies the copying progress for synchronous copy as a percentage. This field has no value (empty field) if the near sites are synchronized.
- ▶ **periodic_copy_progress**
Specifies the copying progress for periodic copy as a percentage. After periodic cycling has finished this value will be *100*.
- ▶ **status**
Specifies the 3-Site relationship status. The value is either *complete* or *partial*. If the value is *partial* check the system for errors while creating the 3-Site replication for this relation.

5.2 Orchestrator 3-Site log files

These two 3-Site Orchestrator services use log files on the Orchestrator as discussed in “Orchestrator services” on page 47:

- ▶ **Task Daemon**
The service **/usr/bin/tstaskd** controls the replication of the configured 3-Site Orchestrator Consistency Groups. The log file of the 3-Site task daemon is:
/opt/ibm/SpectrumVirtualize/rc3site/logs/tstaskd.log
- ▶ **Miscellaneous Daemon**

The service `/usr/bin/tsmiscservicesd` maintains an audit log of all 3-Site Orchestrator commands that are issued. The log file of the 3-Site miscellaneous daemon is:

`/opt/ibm/SpectrumVirtualize/rc3site/logs/tsmiscservicesd.log`

The 3-Site task daemon log files contains error messages and the information on the periodic 3-Site cycling. This information is not listed in the cluster audit log to prevent a confusing cluster audit log due to much messages. There are about 75 entries in the log file for one periodic data cycling of one Consistency Group. Example 5-9 shows a subset of these entries to give an idea on the log file content.

Example 5-9 ttaskd.log example

```
[root@localhost logs]# # some ttaskd.log file lines, shortened view for better readability

13:26:31,578 | run_dc_master_prepare_rccgap_stask:CG0
13:26:33,449 | AP copy complete state prepared
13:26:33,449 | run_dc_auxnear_prepare_rccgap_stask:CG0
13:26:35,319 | AP copy complete state prepared
13:26:35,320 | run_dc_start_remote_copy_stask:CG0
13:26:37,518 | RC Copy complete state: copied
13:26:37,519 | run_dc_start_remote_copy_stask:Waiting for copy complete on far site rightchannel
                    for CG0
13:26:37,833 | RC Copy complete state: copied
13:26:37,833 | run_dc_start_remote_copy_stask:Got copy complete on far
                    site rightchannel for CG0
13:26:37,833 | setup_dc_config_auxfar_prepare_rccgap_stask:CG0
13:26:40,422 | AP copy complete state prepared
13:26:40,422 | setup_dc_config_auxfar_start_rccgap_stask:CG0
13:26:42,642 | AP copy complete state : running
13:26:42,642 | run_dc_mark_copy_complete_stask:CG0
13:26:42,848 | run_dc_rccg_cycle_completion_stask:CG0
```

Example 5-9 shows commands of a periodic data cycling. The Orchestrator uses Spectrum Virtualize **svctask** commands to fulfill the listed tasks and **svcinfo** commands to check the results. Those commands are omitted for better readability. Example 5-10 shows entries after the Near Sites Fibre Channel link is broken.

Example 5-10 Example of log entries when the Near Sites Fibre Channel link is down in a Cascaded configuration

```
[root@localhost logs]# # some ttaskd.log file lines, shortened view for better readability

14:15:15,330 | function condition_check_for_stopped_state:TSCG_SM_STATE_3SC returned True CG0
14:15:15,330 | increment_link_failure_counter TSCG_SM_STATE_STOPPEDCG0 link failure counts
14:15:15,330 | tscg_sm_generate_evt_data_cycle_stopped
14:15:15,331 | running svctask logerror -errnum 330131 -type 95 -id 0 at 10.10.113.247
```

Orchestrator commands changing a configuration, and stopping or starting a 3-Site CG are listed in the log file of the miscellaneous daemon. Example 5-11 shows the **ch3siterccconsistgrp** command used in Example 5-7 on page 51.

Example 5-11 Miscellaneous daemon log entries in tsmiscservicesd.log

```
[root@localhost ]# # some smiscservicesd.log file lines, shortened view for better readability

18:39:15,246 | running at 10.10.113.247
18:39:15,660 | running tscmdlogger -i "ch3siterccconsistgrp -periodicsrc AuxNear CG0"
```

```
-t 1585586355 -r "0" 28 at 10.10.113.247
```

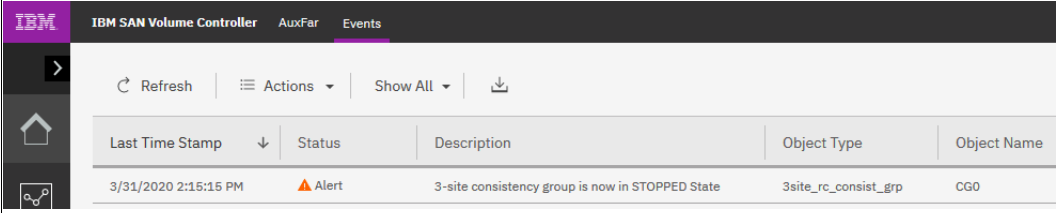
The time stamp in the log file is the time of the Orchestrator.

Note: Time stamps of the Orchestrator log files and of the `ls3sitercconsistgrp` command output may differ because the log file time stamps represent Orchestrator time, and the `ls3sitercconsistgrp` command `freeze_time` represents the AuxFar cluster time.

A good practice is to use Network Time Protocol (NTP) and correct time zones to keep time stamps of the Orchestrator and the cluster synchronized.

5.3 Event logging at the AuxFar site

The event daemon `/usr/bin/eventd` communicates with all three sites and monitors various important attributes of the sites, and the configured Consistency Groups as described in “Orchestrator services” on page 47. The stopped CG described in Example 5-10 on page 53 is shown as an event at the AuxFar site. Figure 5-1 shows this event.



Last Time Stamp	Status	Description	Object Type	Object Name
3/31/2020 2:15:15 PM	Alert	3-site consistency group is now in STOPPED State	3site_rc_consist_grp	CG0

Figure 5-1 CG is in stopped state event

The periodic data cycling is stopped because the periodic source is the secondary volume, and the Metro Mirror between the primary and secondary volume is stopped. After changing the periodic source to the primary volume (on the Master cluster), the periodic data cycling will automatically restart, and the state of the CG will be `2site_periodic` as shown in Example 5-12.

Example 5-12 Changing from Cascade to Star topology

```
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time      cycle_period status
CG0  stopped      Master AuxNear      idling_disconnected stopped      2020/03/31/14/11/40 300      sync_link_offline

[root@localhost ~]# ch3sitercconsistgrp -periodicsrc Master CG0
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time      cycle_period status
CG0  2site_periodic Master Master      idling_disconnected stopped      2020/03/31/15/09/45 300      sync_link_offline
```

The changed configuration and the `2site_periodic` state are shown in the event log at the AuxFar site as shown in Figure 5-2.

IBM SAN Volume Controller AuxFar Events				
↻ Refresh ⋮ Actions Show All ⬇				
Last Time Stamp	Status	Description	Object Type	Object Name
3/31/2020 3:04:28 PM	▲ Alert	3-site consistency group is now in 2-site data cycling mode	3site_rc_consist_grp	CG0
3/31/2020 3:04:28 PM	● Message	Configuration change requested by the last command has completed successfully	3site_rc_consist_grp	CG0
3/31/2020 2:15:15 PM	✓ Alert	3-site consistency group is now in STOPPED State	3site_rc_consist_grp	CG0

Figure 5-2 CG changed to 2site_periodic state

The properties of the event created by the **ch3siterconsistgrp** command are listed in Figure 5-3.

Properties and Sense Data for Event 050988

Configuration change requested by the last command has completed successfully

Mark as Fixed

First Time Stamp

3/31/2020 3:04:28 PM

Last Time Stamp

3/31/2020 3:04:28 PM

Fixed Time Stamp

3/31/2020 3:26:07 PM

Event Count

1

Event ID

050988

Event ID Text

Configuration change requested by the last command has completed successfully

Sequence Number

9000023

Object Type

3site_rc_consist_grp

Object ID

0

Object Name

CG0

Secondary Object ID

Secondary Object Type

Copy ID

Reporting Node ID

2

Reporting Node Name

node2

Root Sequence Number

Error Code

Error Code Text

Dmp Family

IBM

Status

Message

Fixed

Yes

Auto Fixed

Yes

Notification Type

Informational

Previous

Next

Close

Figure 5-3 Completion status of the ch3siterconsistgrp command

Table 5-1 shows the possible event IDs and their event text.

Table 5-1 Event status table

ID	Event	Type
050988	Configuration change requested by the last command has completed successfully	Informational
050990	Configuration change requested by the last command has failed	Warning

ID	Event	Type
050991	The cycle time for 3-Site Consistency Group has exceeded cycle period set by user	Warning
050992	3-Site data cycling is stopped because SVC cluster has lost connection with orchestrator	Warning
050993	3-Site Consistency Group is now in STOPPED State	Warning
050994	3-Site Consistency Group is now in 2-Site data cycling mode	Warning

Stop and start of a Consistency Group

Two commands are used to stop and start a Consistency Group:

- ▶ **stop3sitercconsistgrp** <consistency group>
- ▶ **start3sitercconsistgrp** <consistency group>

The successful execution of the command results in:

- ▶ **stop3sitercconsistgrp CG0**

and is shown at the AuxFar site (shown in Figure 5-4) as event 050988.

Event ID	Last Time Stamp	Status	Description	Object Type	Object Name
050988	4/7/2020 10:50:26 AM	Message	Configuration change requested by the last command has completed successfully	3site_rc_consist_grp	CG0
050993	4/7/2020 10:50:21 AM	Alert	3-site consistency group is now in STOPPED State	3site_rc_consist_grp	CG0
981004	4/7/2020 1:00:04 AM	Message	FC discovery occurred	cluster	AuxFar

Figure 5-4 stop3sitercconsistgrp initiated events at AuxFar site

5.4 Listing 3-Site Consistency Groups using the Master cluster

In a setup with 2-Site and 3-Site Consistency Groups, the Master and AuxNear Remote Copy GUI does not show which Consistency Group is a 3-Site Consistency Group, as shown in Figure 5-5.

Name	State	Master Volume	Auxiliary Volume
CG0	State: Consistent Synchronized	Master System: Master	Auxiliary System: AuxNear
CG1	State: Consistent Synchronized	Master System: Master	Auxiliary System: AuxNear

3-Site tscgl_nawfbf644

Figure 5-5 Two Consistency Groups, only one is 3-Site Consistency Group

The Orchestrator CLI lists the 3-Site Consistency Groups as shown in Example 5-13.

Example 5-13 3-Site Consistency Group list, shortened output

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary
CG0  3site_consistent Master

[root@localhost ~]# ls3sitercrelationship
name  consistgrp master_vdisk_id auxnear_vdisk_id auxfar_vdisk_id
rcrel0 CG0      12             12             0
rcrel1 CG0      13             13             3
```

To list all 3-Site Consistency Groups using the Master CLI, three steps are needed:

1. Use **lsrcrelationship** to find all AccessPoints in a 3-Site relationship by using the 3-Site prefix *tsrell* for a 3-Site relationship.
2. Use **lsfcmap** to find the source volume with the Flash Copy mapping to the AccessPoint.
3. Use **lsrcrelationship** to find the Consistency Group of the source volume.

The three steps are shown in Example 5-14. The two 3-Site relationships to AuxFar are listed with their corresponding AccessPoints, Flash Copy mapping, and the Consistency Group to the AuxNear site.

Example 5-14 Using the Master CLI to find all 3-Site Consistency Groups

```
IBM_FlashSystem:Master:superuser> # find all AccessPoints
superuser>lsrcrelationship -filtervalue "name=tsrell*" -delim : | cut -d: -f6,8
master_vdisk_name:aux_cluster_name
vdisk0:AuxFar
vdisk1:AuxFar

IBM_FlashSystem:Master:superuser> # find all source volumes of the flash copy mappings to the
AccessPoints
superuser>lsfcmap -filtervalue "target_vdisk_name=vdisk0" -delim : | cut -d: -f4,6
source_vdisk_name:target_vdisk_name
VOL_1_L:vdisk0
superuser>lsfcmap -filtervalue "target_vdisk_name=vdisk1" -delim : | cut -d: -f4,6
source_vdisk_name:target_vdisk_name
VOL_2_L:vdisk1

IBM_FlashSystem:Master:superuser> # find the consistenc group of the source volumes.
superuser>lsrcrelationship -filtervalue "master_vdisk_name=VOL_1_L" -delim : | cut -d: -f8,13
aux_cluster_name:consistency_group_name
AuxNear:CG0
superuser>lsrcrelationship -filtervalue "master_vdisk_name=VOL_2_L" -delim : | cut -d: -f8,13
aux_cluster_name:consistency_group_name
AuxNear:CG0
```

5.5 Maintenance

This section describes the planned maintenance tasks. 3-Site Orchestrator provides two commands to alter the configuration of the Orchestrator and of the 3-Site Consistency Groups. These commands are:

► **ch3siteconfig**

The command **ch3siteconfig** changes the current Orchestrator configuration.

► **ch3sitercconsistgrp**

The command **ch3sitercconsistgrp** changes a specified 3-Site remote copy consistency group.

5.5.1 Changing the current Orchestrator configuration

To change the current Orchestrator configuration, complete the task described in the following procedure. Issue the **ch3sitercconsistgrp <attribute> <parameter>** command to change the current Orchestrator configuration.

Note: The command **ch3sitercconsistgrp** fails (except for **-updatesystemname**) if all Consistency Groups are not in a stopped state.

Example 5-15 shows examples on changing the alias name of the master site and the user name.

Example 5-15 ch3sitercconsistgrp examples, command output is shortened

```
# # list the current configuration and stop all consistency groups
[root@localhost rc3site]# ls3siteconfig
master_name Master
. . .
username 3SiteUser
[root@localhost ~]# stop3sitercconsistgrp CG_0
# # rename the master_name site alias
[root@localhost ~]# ch3siteconfig -site Master -sitealias Master_New
# # change the 3-Site username and keyfile
[root@localhost ~]# ch3siteconfig -username 3SiteAdmin

# # list the current configuration
[root@localhost ~]# ls3siteconfig
master_name Master_New
. . .
username 3SiteAdmin

# # rename the master_name site alias to previos name
[root@localhost ~]# ch3siteconfig -site Master_New -sitealias Master
```

Table 5-2 shows the different attributes and parameters.

Table 5-2 *ch3siteconfig* attributes and parameters

Attribute	Parameter	Description
-keyfile	ssh_keyfile_path	(Optional) Specifies the updated path for the private SSH key file for the 3-Site user on the host. This parameter is mutually exclusive with -site.
-username	3site_username	(Optional) Specifies the updated 3Site username for all three sites. This parameter is mutually exclusive with -site.
-port	local_port_offset	(Optional) Specifies the updated starting address for the three local ports that are used for connection with the three sites over the SSH tunnel.
-site <site_name> <i>site_name</i> specifies the site to update	-ip1 site_ip1	(Optional) Specifies the new IPv4 address for IP address 1. This parameter is mutually exclusive with -ip2.
-site <site_name> <i>site_name</i> specifies the site to update	-ip2 site_ip2	(Optional) Specifies the new IPv4 address for IP address 2. This parameter is mutually exclusive with -ip1.
-site <site_name> -sitealias <i>site_name</i> specifies the site to update	new_site_alias	(Optional) Specifies the updated site alias name.
-updatesystemname		(Optional) Specify this parameter to refresh the system name change of any of the three site

5.5.2 Changing the cycle period

To change the cycle period of a consistency group, complete the task described in the following procedure. Issue the **ch3sitercconsistgrp -cycleperiod** command to change the cycle period of a consistency group as shown in Example 5-16.

Example 5-16 Changing the cycle period to the minimum of 300 seconds (command output shortened)

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source cycle_period status
CG_0 3site_consistent Master Master          500          online
[root@localhost ~]# ch3sitercconsistgrp -cycleperiod 300 CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source cycle_period status
CG_0 3site_consistent Master Master          300          online
```

5.5.3 Changing the periodic source

Before changing the periodic source these two prerequisites should be considered:

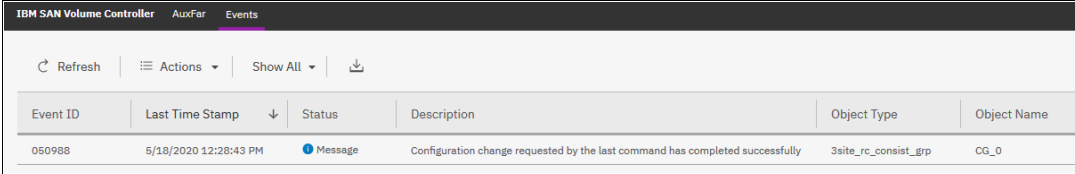
1. Change the periodic source only in the 3site_consistent, 3site_inconsistent, and stopped states.
2. Changing the periodic source to a non-primary site is not supported if the Metro Mirror relationship between the primary site and the AuxNear site is not active.

Issue the **ch3sitercconsistgrp -periodicsrc** command to change the periodic source of a consistency group as shown in Example 5-17.

Example 5-17 Changing the period source (command output shortened)

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 3site_consistent Master Master          consistent_synchronized
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc AuxNear CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 3site_consistent Master AuxNear          consistent_synchronized
```

The configuration request is monitored at the AuxFar site as described in 5.3, “Event logging at the AuxFar site” on page 54. The command completed successfully notification event is shown in Figure 5-6.



Event ID	Last Time Stamp	Status	Description	Object Type	Object Name
050988	5/18/2020 12:28:43 PM	Message	Configuration change requested by the last command has completed successfully	3site_rc_consist_grp	CG_0

Figure 5-6 Event monitoring at the AuxFar site.

5.5.4 Changing the primary site

Before changing the primary site these two prerequisites should be considered:

1. The application must be in the stopped state on the current primary site. All volume mappings from the primary site to the application must be removed.
2. Perform this procedure in the 3site_consistent, 3site_inconsistent, and stopped states when the Metro Mirror relationship between the primary site and the AuxNear site is operational. As a result, the sync_copy_state value is consistent_synchronized.

Issue the **ch3sitercconsistgrp -primary** command to change the primary site of a Consistency Group as shown in Example 5-18.

Example 5-18 Changing the primary site and restarting the CG (command output shortened)

```
[root@localhost ~]# stop3sitercconsistgrp CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 stopped Master AuxNear          consistent_synchronized
[root@localhost ~]# ch3sitercconsistgrp -primary AuxNear CG_0
[root@localhost ~]# start3sitercconsistgrp CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 3site_consistent AuxNear AuxNear          consistent_synchronized
```

The configuration request is monitored at the AuxFar site as described in 5.3, “Event logging at the AuxFar site” on page 54. The command notification event is shown in Example 5-19.

Example 5-19 Orchestrator Event

```
IBM_2145:AuxFar:superuser>ls7eventlog | tail -1
9000029 200518124045 3site_rc_consist_grp 0 CG_0 message no 050988
Configuration change requested by the last command has completed successfully
```


The changed primary site is shown in the *Master* GUI as shown in Figure 5-7.

Name	State	Master Volume	Auxiliary Volume
Not in a Group			
CG_0	State: Consistent Synchronized	Master System: Master	Auxiliary System: AuxNear
rcrel0	Consistent Synchronized	Master and Auxiliary are Switched	VOL_1_R
rcrel1	Consistent Synchronized	V_2_L	VOL_2_R

Figure 5-7 Event monitoring at the AuxFar site.

5.5.5 Performing planned maintenance on a primary site or auxiliary-near site

A planned maintenance, for example, a service action that will take one site offline, needs to consider these three prerequisites:

1. To exclude a site for maintenance, the site must not be the primary or periodic-source site.
2. Exclusion of the site from data cycling stops the Metro Mirror Consistency Group between the two near sites.
3. Every Consistency Group has to be excluded.

The primary and periodic-source site is still replicating the data to the auxiliary-far site if the other near site is excluded from the data cycling. This assures an asynchronous data copy while one near site is excluded.

To exclude a site from 3-Site replication, complete the task described in the following procedure for every 3-Site Consistency Group. Issue the **ch3sitercconsistgrp -exclude** command to exclude a Consistency Group as shown in Example 5-20.

Example 5-20 Excluding the Master site (command output shortened)

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary sync_copy_state      periodic_copy_state status
CG_0 3site_consistent AuxNear consistent_synchronized stopped             online
[root@localhost ~]# ch3sitercconsistgrp -exclude Master CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary sync_copy_state      periodic_copy_state status
CG_0 2site_periodic AuxNear consistent_stopped stopped             sync_link_stopped
```

The state of the Consistency Group is now 2site_periodic and the link status between the two near sites is stopped after the **ch3sitercconsistgrp -exclude** command has successfully finished. The command notification event is shown Example 5-21.

Example 5-21 Orchestrator Event

```
IBM_2145:AuxFar:superuser>ls7eventlog | tail -1
9000031 200518125948 3site_rc_consist_grp 0 CG_0 alert no 050994
3-site consistency group is now in 2-site data cycling mode
```

The excluded site can be included after the site maintenance using the **ch3sitercconsistgrp -join** command and parameter as shown in Example 5-22 on page 62.

Example 5-22 Joining the Master site (command output shortened)

```
[root@localhost ~]# ch3sitercconsistgrp -join Master CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary sync_copy_state      periodic_copy_state status
CG_0 3site_consistent AuxNear consistent_synchronized stopped              online
```

The state of the Consistency Group will be 3site_consistent and the link status between the two near sites will be online after the **ch3sitercconsistgrp -join** command has successfully finished and the data between both near sites is replicated and consistent. The command notification event is shown in Example 5-23.

Example 5-23 Orchestrator Event

```
IBM_2145:AuxFar:superuser>lsevenlog | tail -1
9000032 200518153139 3site_rc_consist_grp 0 CG_0 message no 050988
Configuration change requested by the last command has completed successfully
```

5.5.6 Performing planned maintenance on an AuxFar site

For a planned maintenance procedure, for example, a service action that will take one site offline, follow the steps described in this procedure.

1. Stop 3-Site data cycling between the near and far sites for each Consistency Group.
2. Perform the AuxFar maintenance tasks.
3. After completion of the AuxFar maintenance, start 3-Site data cycling between the near and far sites for each Consistency Group.

To perform planned maintenance at the AuxFar site only, data cycling to this site has to be stopped using the **stop3sitercconsistgrp** command for every 3-Site Consistency Group. The mirroring between the two near sites is not affected and is ongoing as shown in Example 5-24.

Example 5-24 AuxFar maintenance steps

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 3site_consistent AuxNear AuxNear              consistent_synchronized
[root@localhost ~]# stop3sitercconsistgrp CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 stopped AuxNear AuxNear              consistent_synchronized

[root@localhost ~]# # AuxFar maintenance tasks

[root@localhost ~]# start3sitercconsistgrp CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state
CG_0 3site_consistent AuxNear AuxNear              consistent_synchronized
```

5.6 Upgrading the storage system software

Before upgrading the storage system software in a 3-Site configuration of Spectrum Virtualize clusters, the following two prerequisites should be considered:

1. Only one Spectrum Virtualize cluster should be upgraded at a time.
2. Replication to the AuxFar site has to be stopped for all 3-Site Consistency Groups.

To upgrade the storage system, data cycling to the auxiliary-site has to be stopped using the **stop3sitercconsistgrp** command for every 3-Site Consistency Group as shown in Example 5-25, and restarted after the upgrade.

Example 5-25 Upgrading a storage system

```
[root@localhost ~]# stop3sitercconsistgrp CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state
CG_0 stopped Master  Master                  consistent_synchronized

[root@localhost ~]# # Upgrade of a spectrum virtualize storage system

[root@localhost ~]# start3sitercconsistgrp CG_0
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state
CG_0 3site_consistent Master  Master                  consistent_synchronized
```

The procedure described in Example 5-25 has to be repeated for every cluster. This assures replication to the auxiliary-site after every cluster software upgrade, and minimizes the lack of the RPO caused by the stopping of the 3-Site replication during a storage system upgrade.



Failure protection and recovery procedures for 3-Site Replication

This chapter discusses different failure scenarios, how to handle them using the three site capabilities, and the current limits and restrictions. It contains the following scenarios:

- ▶ Link failures between sites
- ▶ Single and multiple site or storage system failures
- ▶ Loss of the Orchestrator

A loss of a site, the link between those sites, or a storage system can be a temporary or permanent issue. In any case, a manual decision whether a site change is appropriate or not is required.

In this chapter we show how to perform 3-Site recovery operations using the 3-Site Orchestrator. We do not discuss the additional, required steps to enable host access and get those volumes back online. Those topics are discussed in great detail in:

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1, SG24-8465*
- ▶ *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1, SG24-8466*
- ▶ *Implementing the IBM FlashSystem 5010 and FlashSystem 5030 with IBM Spectrum Virtualize V8.3.1, SG24-8467*

6.1 General failure handling

Maintaining a warm site with a copy of your data ready-to-go is the usual way of ensuring rapid recovery from a large-scale failure. Adding a third site provides an additional data protection level for protection against multiple site failures.

In any of those disaster scenarios, manual intervention is required for providing data access to the appropriate hosts, changing copy directions, or declaring a site as temporarily or permanently lost. The required steps depend on the existing topology and failure scenario. To provide consistency you should perform all the required 3-Site changes on the 3-Site Orchestrator.

Note: It is recommended to contact IBM support for 3-Site solution recovery assistance.

6.2 Recovery from link failures

Link failures between the sites can be due to temporary or permanent issues. In the case of a permanent issue, a topology change is potentially required.

The *Active Periodic Link* describes the active link used for data replication from one near site to the far site.

The *Inactive Periodic Link* describes the already configured link to the AuxFar site, normally not used for data replication, both are shown as an example in Figure 6-1.

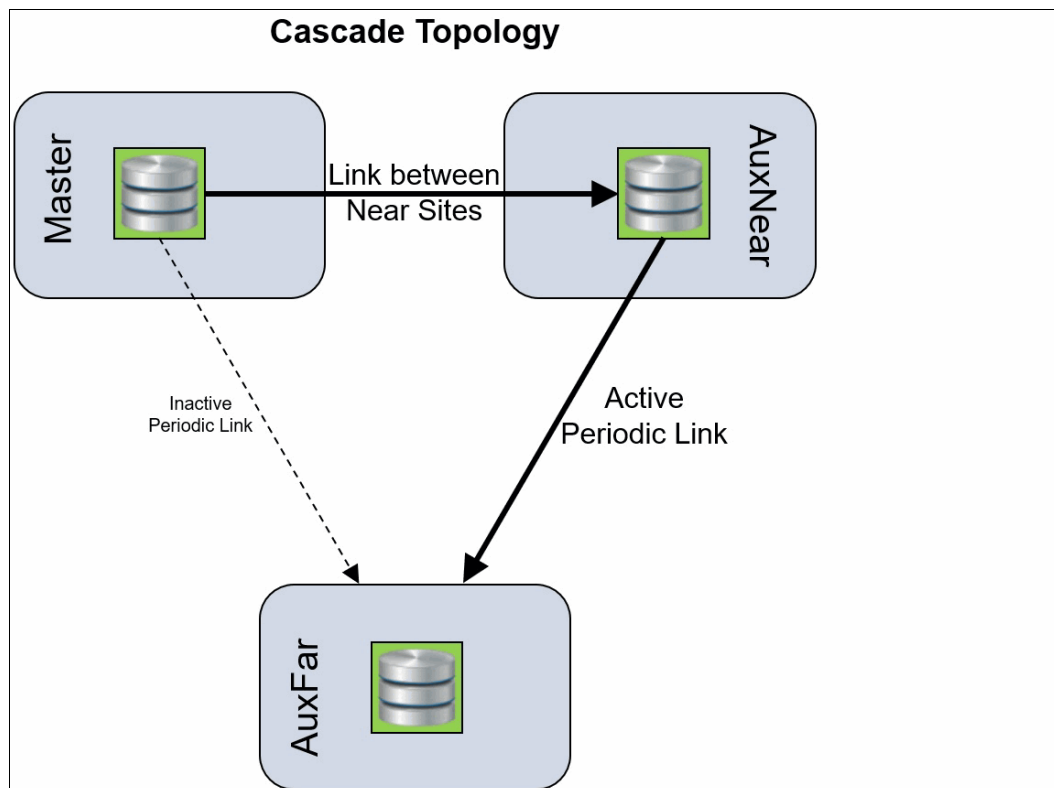


Figure 6-1 Link description between main sites

6.2.1 Link failure between near sites

If data replication between both near sites has failed, you must assume that the AuxNear target is not getting any updates any longer, and will stay on the last consistent data copy before the link failure occurred.

In a Cascade topology, the AuxFar replication depends on the successful replication between both near sites, and so we must differentiate between Star and Cascade topologies.

Star topology

To recover from a link failure between near sites in a Star topology as shown in Figure 6-2 on page 67, complete the steps described in this procedure.

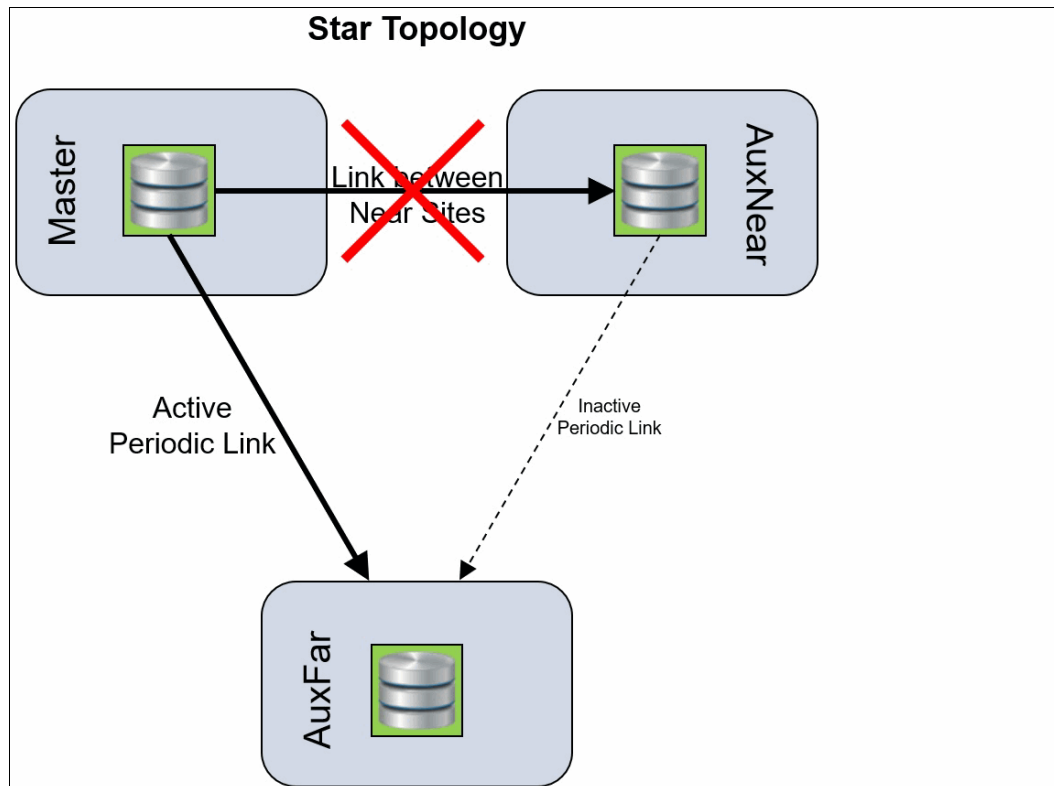


Figure 6-2 Link failure between near sites in a Star topology

1. Verify that the 3-Site Consistency Group is now in 2-Site data cycling mode. Verify the event notification using the event log.

Issue the **ls3sitercconsistgrp** Orchestrator CLI command to verify the state of the Consistency Groups. In the resulting output, the state of the Consistency Groups will change after a while to `2site_periodic` as shown in Example 6-1.

Example 6-1 Verify state of Consistency Group

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state      periodic_copy_state
freeze_time        cycle_period status
CG0 3site_consistent Master Master          consistent_synchronized stopped
2020/04/01/14/08/54 300          online

[root@localhost ~]# ls3sitercconsistgrp
```

```

name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 2site_periodic Master Master          idling_disconnected stopped
2020/04/01/14/11/44 300          sync_link_offline

```

The Orchestrator sends the status change to the AuxFar cluster, as described in Chapter 5, “3-Site Replication Monitoring and Maintenance” on page 45. To verify, in the AuxFar cluster GUI select **Monitoring** → **Events** to list the events. The last event shows the event text “3-Site Consistency Group is now in 2-Site data cycling mode” as shown in Figure 6-3.

Last Time Stamp	Status	Description	Object Type	Object Name
4/1/2020 3:11:38 PM	Alert	3-site consistency group is now in 2-site data cycling mo...	3site_rc_consist_grp	CG0

Figure 6-3 Event 2site_periodic periodic cycling

2. Repair the link between the near sites.
3. Issue the **ls3sitercconsistgrp** command again to verify the state of the Consistency Groups.
4. If the `sync_copy_state` of a CG is `consistent_stopped`, issue the **ch3sitercconsistgrp -join** command to join the site in 3-Site data cycling and verify the repair action. Use the site alias defined when the 3-Site configuration was created. This name can be listed using the **ls3siteconfig** command. This step is shown in Example 6-2.

Example 6-2 Join the site in 3-Site data cycling

```

[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 2site_periodic Master Master          consistent_stopped stopped
2020/04/01/15/32/09 300          sync_link_stopped

[root@localhost ~]# ls3siteconfig | grep auxnear_name
auxnear_name AuxNear
[root@localhost ~]# ch3sitercconsistgrp -join AuxNear CG0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 3site_consistent Master Master          consistent_synchronized stopped
2020/04/01/15/36/59 300          online

```

5. Finally, *Run fix* procedures against event log entries at the Master site and the AuxFar site. Figure 6-4 shows the Master cluster GUI with the unfixed event.

IBM FlashSystem 9100 Master Events					
Recommended Action: ✖ Error 1720 : Remote Copy - lost synchronization Run Fix					
↻ Refresh ⋮ Actions Show All ⬇ Download					
Error Code	Last Time Stamp	Status	Description	Object Type	Object Name
	4/1/2020 3:36:54 PM	📢 Message	Remote Copy complete	rc_consist_grp	CG0
	4/1/2020 3:28:39 PM	📢 Message	Remote Copy ready to start	rc_consist_grp	CG0
1720	4/1/2020 3:12:04 PM	✖ Alert	Remote Copy - lost synchronization	rc_consist_grp	CG0

Figure 6-4 The lost Metro Mirror synchronization has to be fixed manually

Cascade topology

In a Cascade topology the status and the RPO of the AuxFar replication depends on the successful replication between both near sites as shown in Figure 6-5. There are two options as to how to proceed, if the link between the two near sites has failed.

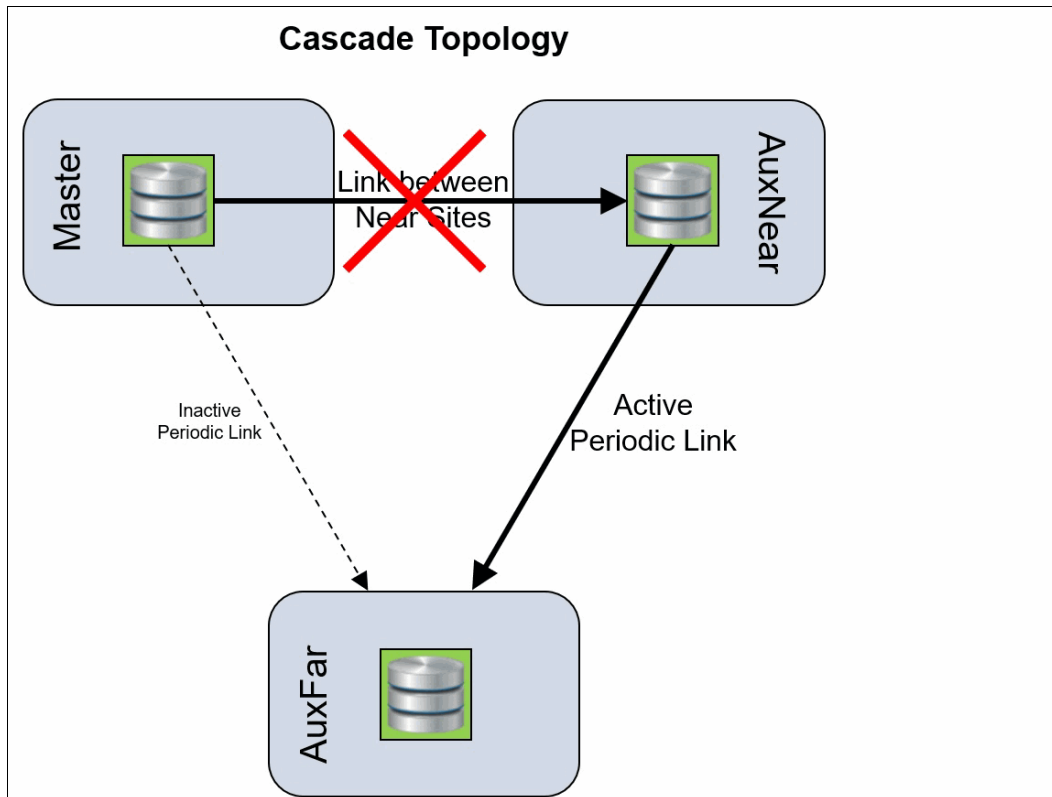


Figure 6-5 Link failure between near sites in a cascaded topology

► Option 1: temporarily change to a star topology

Temporarily change to a star topology which enables replication between the Master and the AuxFar site. This option provides an up-to-date second (asynchronous) data copy although the synchronous mirror between the near sites is not available. The 3-Site mirror must be changed to a 2site_periodic state system. A switch back to the former cascade topology is possible as soon as the link between both near sites is repaired.

The procedure is described in 6.2.2, "Procedure to temporarily change to a star topology" on page 70.

► Option 2: rejoin the site to a 3-Site configuration

Keep the current topology and re-establish the connection between both near sites as soon as the link between those two sites is back to normal operation. The data copy on AuxNear and AuxFar cannot be updated until the link between both near sites is back to normal operation mode. So, there is no up-to-date data copy available on both sites and you directly rejoin the site back to the 3-Site configuration.

The steps required for this operation are described in 6.2.3, “Procedure to rejoin the site to a 3-Site configuration” on page 71.

6.2.2 Procedure to temporarily change to a star topology

To perform this task, complete the following steps:

1. Take immediate action and move the Consistency Group to the `2site_periodic` state by issuing the `ch3sitercconsistgrp -periodicsrc` command. Use the `ls3sitercconsistgrp` command to verify the state of the Consistency Groups as shown in Example 6-3.

Example 6-3 Move to Consistency Group to 2site_periodic state

```
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc Master CG0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 2site_periodic Master Master              idling_disconnected stopped
2020/04/01/17/09/19 300          sync_link_offline
```

2. A reconfiguration to the previous setup is required as soon as the link between both near sites is back to normal operation. The resynchronization process may takes some time, Monitor the process until the next periodic cycling is completed to get back to the `3site_consistent` status as shown in Example 6-4. The difference to the previous command output is highlighted for better readability.

Example 6-4 Rejoin the AuxNear system

```
[root@localhost ~]# # link repaired
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 2site_periodic Master Master              consistent_stopped stopped
2020/04/01/17/31/14 300          sync_link_stopped
```

```
[root@localhost ~]# ch3sitercconsistgrp -join AuxNear CG0

[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 2site_periodic Master Master              inconsistent_copying stopped
2020/04/01/17/33/14 300          online
```

```
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state    periodic_copy_state
freeze_time        cycle_period status
CG0 2site_periodic Master Master              consistent_synchronized stopped
2020/04/01/17/33/14 300          online
```

```
[root@localhost ~]# ls3sitercconsistgrp
```

```

name state          primary periodic_copy_source sync_copy_state      periodic_copy_state
freeze_time        cycle_period status
CG0 3site_consistent Master Master          consistent_synchronized stopped
2020/04/01/17/38/25 300          online

```

3. Run `fix` procedure at the event log of the master system as shown in Figure 6-6.

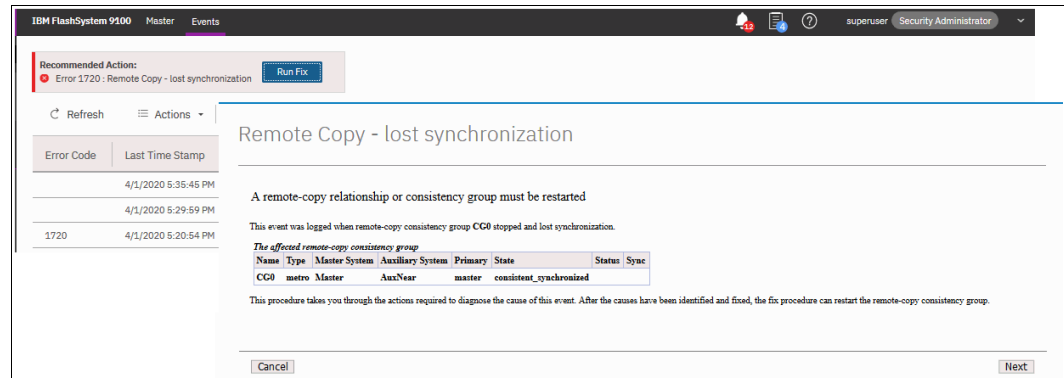


Figure 6-6 Recommended Action: Run Fix: Remote Copy - lost synchronization

4. Change the setup back to a cascade topology and verify the configuration.

Example 6-5 Change the configuration back to cascade topology and verify

```
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc AuxNear CG0
```

```

[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state      periodic_copy_state
freeze_time        cycle_period status
CG0 3site_consistent Master AuxNear          consistent_synchronized copying
2020/04/01/18/08/25 300          online

```

6.2.3 Procedure to rejoin the site to a 3-Site configuration

Recover from the periodic state, or the stopped state, without moving into the 2site_periodic state.

After restoring the link failure between the near sites, complete the following steps:

1. Verify that the 3-Site Consistency Group is now in the stopped state and that the AuxFar cluster has received the event log notification as shown in Figure 6-7.

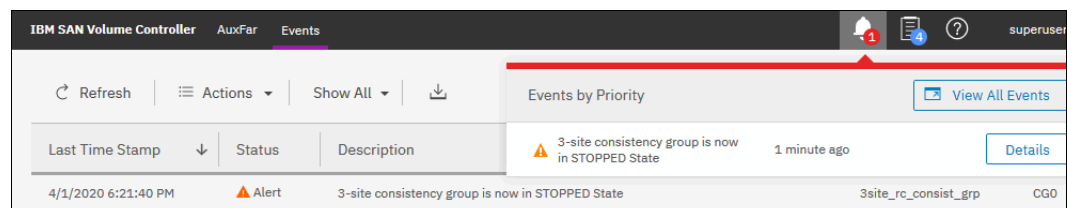


Figure 6-7 CG is in stopped state

2. Issue the `ls3sitercconsistgrp` command to verify the state of the Consistency Groups. In the resulting output, the state of the Consistency Groups is stopped. The link is restored

and therefore the 3-Site data cycling status is `sync_link_stopped` as shown in Example 6-6.

Example 6-6 Verify state of Consistency Group

```
[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time
cycle_period status
CG0 stopped Master AuxNear consistent_stopped stopped
2020/04/01/18/20/45 300 sync_link_stopped
```

If the `sync_copy_state` is `consistent_stopped`, and the state of the 3-Site Consistency Group does not change to `3site_consistent`, complete the following steps:

3. Issue the **ch3sitercconsistgrp -join** command to recover from the stopped state without entering the state `2site_periodic`.
4. Use the **ls3sitercconsistgrp** command to verify the state of the Consistency Groups. The state change might take time and is only reflected after the `sync_copy_state` changes to `consistent_synchronized` as shown in Example 6-7. The state change might take time, several status must be expected during that process as shown in Example 6-7.

Example 6-7 Recover from periodic or stopped state

```
[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time
cycle_period status
CG0 stopped Master AuxNear consistent_stopped stopped
2020/04/01/18/20/45 300 sync_link_stopped

[root@localhost ~]# ch3sitercconsistgrp -join AuxNear CG0

[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state
freeze_time cycle_period status
CG0 3site_consistent Master AuxNear consistent_synchronized stopped
2020/04/01/18/55/40 300 online
```

5. Run *fix* procedure on the event log entry at the master system as shown in Figure 6-7 on page 71.

6.2.4 Link failure of the Active Periodic link

The Active Periodic Link connects the AuxFar site with the replication source site as shown in a Star topology in Figure 6-8 on page 73. To recover from an active-periodic link failure, complete the steps described in this procedure.

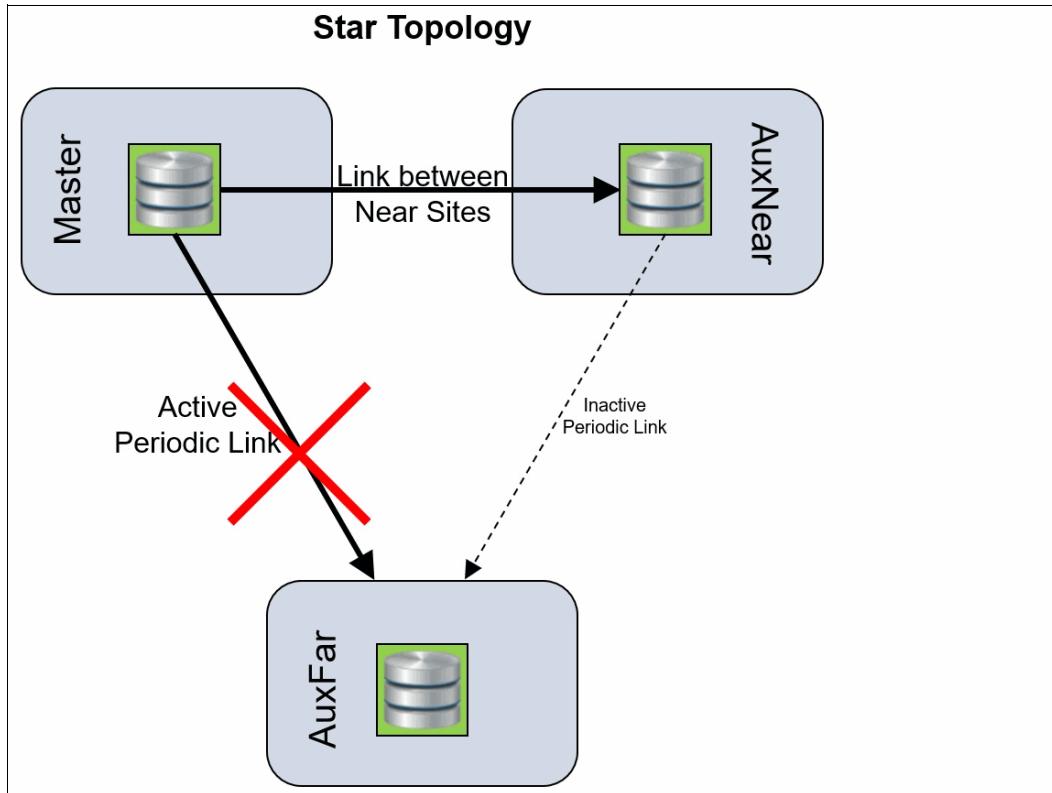


Figure 6-8 Active Periodic Link failure in a Star topology

1. Verify that the 3-Site Consistency Group is now in a stopped state and that the AuxFar cluster has received the event log notification as shown in Figure 6-9. The stopped state and the lost connection are shown.

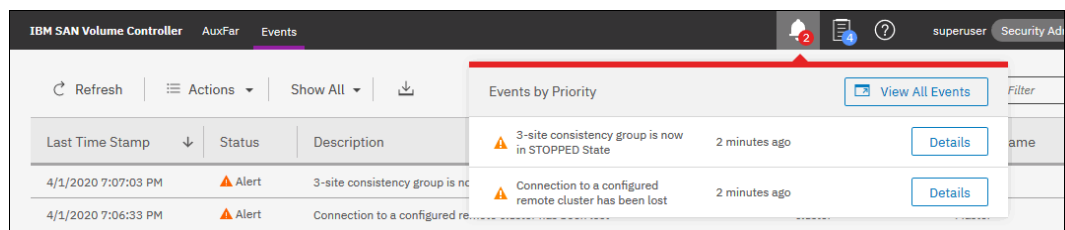


Figure 6-9 CG is in stopped state

2. Issue the `ls3sitercconsistgrp` command to verify the state of the Consistency Group. In the resulting output, the state of the Consistency Group is stopped as shown in Example 6-8.

Example 6-8 Verify state of the Consistency Group

```
[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state
freeze_time cycle_period status
CG0 stopped Master Master consistent_synchronized stopped
2020/04/01/19/06/10 300 active_link_offline
```

3. To complete the link failure recovery, use one of the following options:

- a. Restore the failed link and data cycling automatically restarts. The 3-Site alert event at the AuxFar cluster is automatically fixed as shown in Figure 6-10, and the event concerning the lost connection to the remote cluster has to be fixed manually.

Last Time Stamp	Status	Description	Object Type	Object Name
4/1/2020 7:07:03 PM	✓ Alert	3-site consistency group is now in STOPPED State	3site_rc_consist_grp	CG0
4/1/2020 7:06:33 PM	⚠ Alert	Connection to a configured remote cluster has been lost	cluster	Master

Figure 6-10 Fixed 3-Site

- b. Switch the periodic replication source
- Use the **ch3sitercconsistgrp -periodicsrc** command to switch the source location. The state will be `3site_consistent` and the status will be `inactive_link_offline` as shown in Example 6-9.

Example 6-9 Switch periodic replication source

```
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc AuxNear CG0
[root@localhost ~]# ls3sitercconsistgrp
name state          primary periodic_copy_source sync_copy_state      periodic_copy_state
freeze_time        cycle_period status
CG0 3site_consistent Master AuxNear      consistent_synchronized stopped
2020/04/02/10/41/57 300      inactive_link_offline
```

- Restore the link and use the command **ch3sitercconsistgrp -periodicsrc** again to switch the active periodic replication source back to the original link.

Example 6-10 Switch periodic replication source

```
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc Master CG0
```

6.2.5 Link failure of the Inactive Periodic Link

The Inactive Periodic Link connects the AuxFar site with the replication source site as shown in a Star topology in Figure 6-11 on page 75.

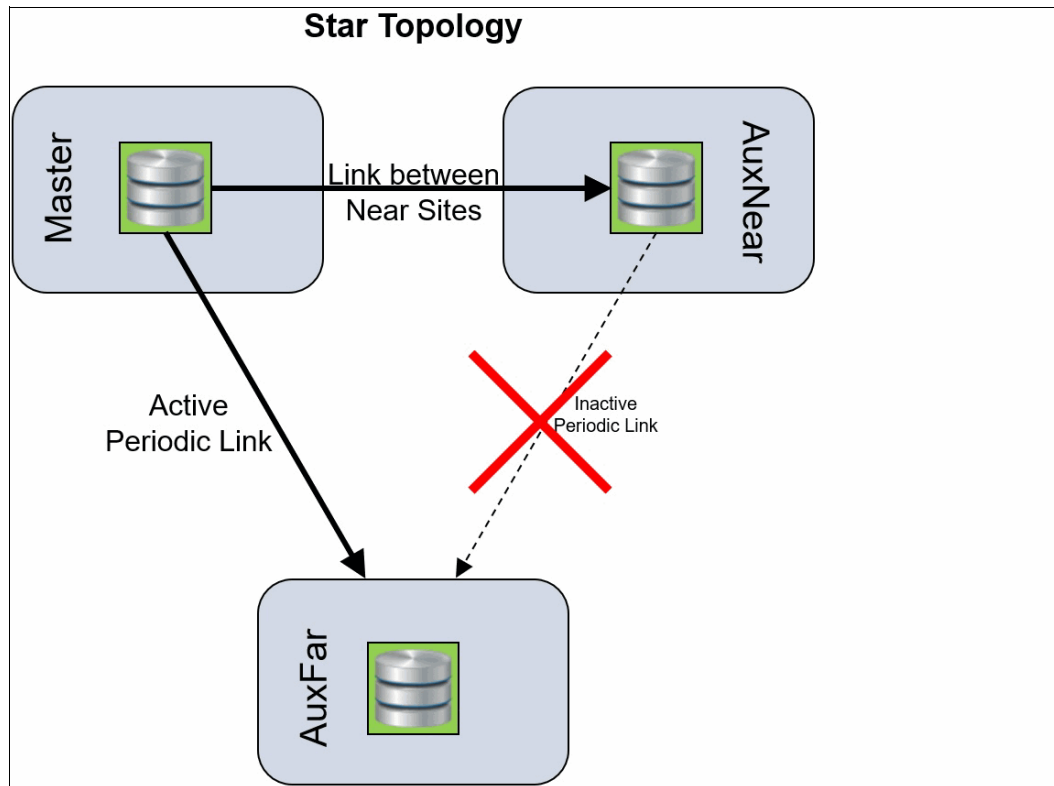


Figure 6-11 Inactive Periodic Link failure in a Star topology

To recover from an inactive periodic link failure, complete the following steps:

1. Issue the `ls3sitercconsistgrp` command to verify the state of the Consistency Groups. In the resulting output, the status is `inactive_link_offline` as shown in Example 6-11.
2. Restore the link at the earliest opportunity.

Example 6-11 Verify the state of the Consistency Group

```
[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time
cycle_period status
CG0 3site_consistent Master Master consistent_synchronized idling 2020/03/25/17/38/47 300
inactive_link_offline
```

After the inactive periodic link is back online, the error will clear automatically.

6.3 Recovery from a site loss or a storage failure

A site loss or a storage system failure can be temporary or permanent. In either case, manual interaction is usually required for recovery.

Note: This section does not cover the steps required to fix the storage system itself or recover host application operation. It focuses only on the recovery of 3-Site Replication operations. For information on storage system and host recovery, contact IBM support or refer to the following publications: *IBM System Storage SAN Volume Controller and Storwize V7000 Replication Family Services*, SG24-7574, *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1*, SG24-8466.

In a 3-Site replication solution, a site storage failure implies a situation when a storage system itself is up and running, can be accessed from the 3-Site Orchestrator and can detect both its partners over replication links, *but* has a failure that takes a volume, which is a member of the 3-Site replication relationship, offline. An example of such a kind of a failure is a storage pool offline situation.

A site failure can also mean a situation when a storage system on a site is not visible through both replication links and the 3-Site Orchestrator SSH link. For example, the site failure may be caused by a power outage that takes the entire site offline. Also, an auto-recovered cluster-wide disaster (Tier 2 (T2) recovery or automatic cluster recovery) is an example of a site failure.

A special case of a site failure is the case of a cluster that has undergone a Tier 3 (T3) recovery, or system recovery. During T3 recovery, inter-system partnerships and relationships are not restored and must be re-created manually. IBM Support must be involved to restore 3-Site Replication operations after a cluster on one of the sites is recovered using T3 procedures. Post T3 recovery steps are intentionally not covered in this book.

6.3.1 Determining the disaster scope and the recovery plan

Recovery steps depend on the 3-Site Replication topology and the primary site assignment at the time of the disaster. You can use the **ls3sitercconsistgrp** command to determine 3-Site Replication state and status as shown in Example 6-12.

Example 6-12 Example of an output for a CG in an online state

```
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state      periodic_copy_state freeze_time      cycle_period status
CG0 3site_consistent Master AuxNear      consistent_synchronized copying      2020/04/01/18/08/25 300      online
```

Here, the *primary* column indicates the application primary site, and *periodic_copy_source* shows which site is a source for replication to AuxFar. If primary and periodic_copy_source match, 3-Site replication is running in a Star topology.

In case of a disaster on a site, the Consistency Group *status* will change from *online* to one of those shown in Table 6-1.

Table 6-1 Consistency Group status

status	Description
primary_offline	Primary site for this Consistency Group is down or it is isolated from the partnership (there is no available network and the site is unable to be reached from the remaining two sites).
primary_storage_offline	Primary site for this Consistency Group is available, but one or more primary site volumes in this CG are offline.

status	Description
primary_and_periodic_src_offline	CG is configured to Star topology (primary and periodic source site match), and Primary site is down or isolated.
periodic_src_offline	Non-primary near site which is a periodic source is down or isolated. (Cascade topology)
periodic_src_storage_offline	Non-primary near site which is a periodic source is available but one or more its volumes in this CG are offline. (Cascade topology)
partner_offline	Near site which is non-primary and not a periodic source is down or isolated. (Star topology)
partner_storage_offline	Near site which is non-primary and not a periodic source is available, but one more its volumes in this CG are offline. (Star topology)
auxfar_offline	System on the AuxFar site is down or isolated.
auxfar_storage_offline	System on the AuxFar site is available but one or more volumes of this CG are offline.

Figure 6-12 summarizes the recovery actions for each kind of site failure. It shows the *state* and *status* that the Consistency Group may get into in case of a disaster at one of the sites.

#	Primary	Topology	Failure at	CG state	CG status	Actions
#1	Master	Star	AuxFar	Stopped	auxfar_offline auxfar_storage_offline	1. Recover AuxFar 2. Start cycling if required
#2			AuxNear	2site_periodic	partner_offline, partner_storage_offline	1. Recover AuxNear 2. Rejoin AuxNear site if required
#3			Master	Stopped	primary_and_periodic_src_offline	1. If required, use AuxNear CLI or GUI to switch MM primary. 2. If required, switch periodic source to run in 2site_periodic mode 3. Recover and rejoin Master site
#1		Cascade	AuxFar	Stopped	auxfar_offline auxfar_storage_offline	1. Recover AuxFar 2. Start cycling if required
#4			AuxNear	Stopped	periodic_src_offline periodic_src_storage_offline	1. Switch periodic source 2. Recover AuxNear and switch source back to it
#5			Master	Stopped	primary_offline primary_storage_offline	1. If required, use AuxNear CLI or GUI to switch MM primary. 2. Recover and rejoin Master site

Figure 6-12 Site failure types and recovery actions

All listed scenarios are explained with more detail in the following sections.

6.3.2 Recovery from a disaster at the AuxFar site (scenario #1)

In both site failure and storage failure cases at the AuxFar site, independently of a 3-Site topology, recovery steps remain the same. Figure 6-13 shows an example of a disaster which takes down the AuxFar site in a cascade topology.

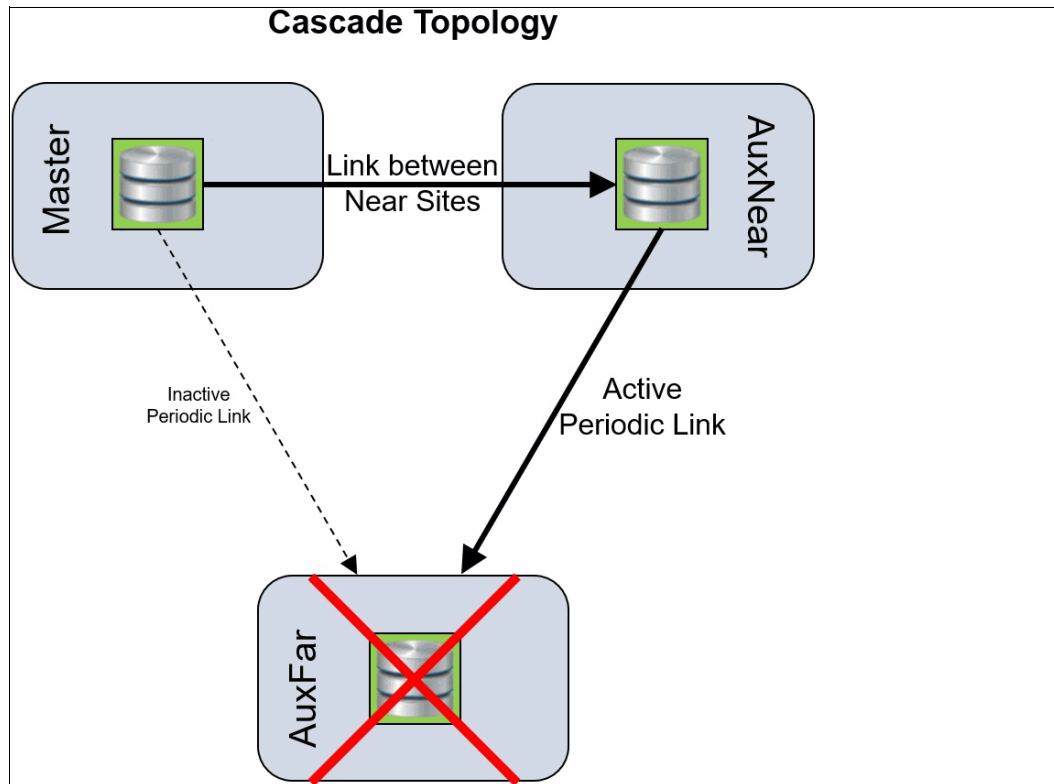


Figure 6-13 AuxNear site failure in a Cascade topology

1. Issue the `ls3sitercconsistgrp` command to verify the state of the Consistency Groups. In case of a failure on AuxFar, the state of the Consistency Group is *stopped*, and its status can be *auxfar_storage_offline* or *auxfar_offline*, as shown in Example 6-13.

Example 6-13 AuxFar offline example

```
[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state freeze_time cycle_period status
CG0 stopped Master AuxNear consistent_synchronized stopped 2020/04/02/15/52/12 300 auxfar_offline
```

2. Perform the required disaster recovery procedures to recover the storage system on the AuxFar site to a fully operational state.
3. After the AuxFar site is recovered and 3-Site Orchestrator access is restored, data cycling will start automatically.
4. If a storage failure still exists after bringing the volume online, start data cycling manually with the `start3sitercconsistgrp` command.

6.3.3 Recovery from a disaster at non-primary AuxNear site in a star topology (scenario #2)

An example of a non-primary site failure in a star topology is shown in Figure 6-14 on page 79.

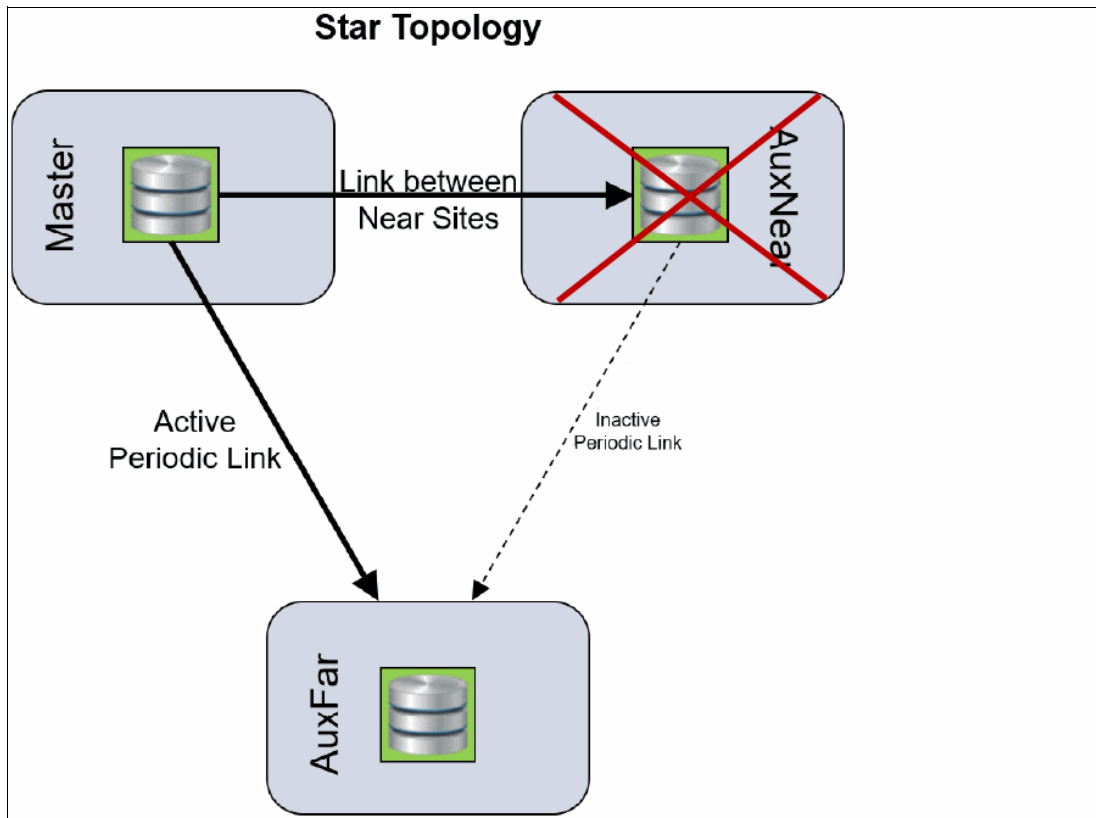


Figure 6-14 AuxNear failure example

To recover, follow these steps.

1. Issue the **ls3sitercconsistgrp** command to verify the state of the Consistency Group. In the resulting output, the state of the Consistency Group is `2site_periodic`, as shown in Example 6-14.

Example 6-14 Verify state of the Consistency Groups

```
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state  periodic_copy_state freeze_time  cycle_period status
CG0 2site_periodic Master Master      idling_disconnected stopped          2020/04/02/16/47/37 300      partner_offline
```

2. Perform recovery activities on the AuxNear site to restore it to a fully functional state.
3. To complete recovery of the 3-Site setup, check `sync_copy_state` in **ls3sitercconsistgrp** output:
 - If it is `consistent_synchronized`, the site is automatically included in data cycling.
 - If it is `consistent_stopped`, issue the **ch3sitercconsist -join** command to include the site in data cycling, as shown in Example 6-15.

Example 6-15 Joining a site

```
[root@localhost ~]# ch3sitercconsistgrp -join AuxNear CG0
```

4. Issue the **ls3sitercconsistgrp** command to verify that recovery is complete, as shown in Example 6-16.

Example 6-16 Verify the recovery

```
[root@localhost ~]# ls3sitercconsistgrp
name state      primary periodic_copy_source sync_copy_state  periodic_copy_state freeze_time  cycle_period status
CG0 2site_periodic Master Master      idling_disconnected stopped          2020/04/02/16/47/37 300      partner_offline
```

6.3.4 Recovery from a disaster on a non-primary AuxNear site in a cascade topology (scenario #4)

An example of a non-primary site failure in a cascade topology is shown in Figure 6-15. The recovery steps are discussed below.

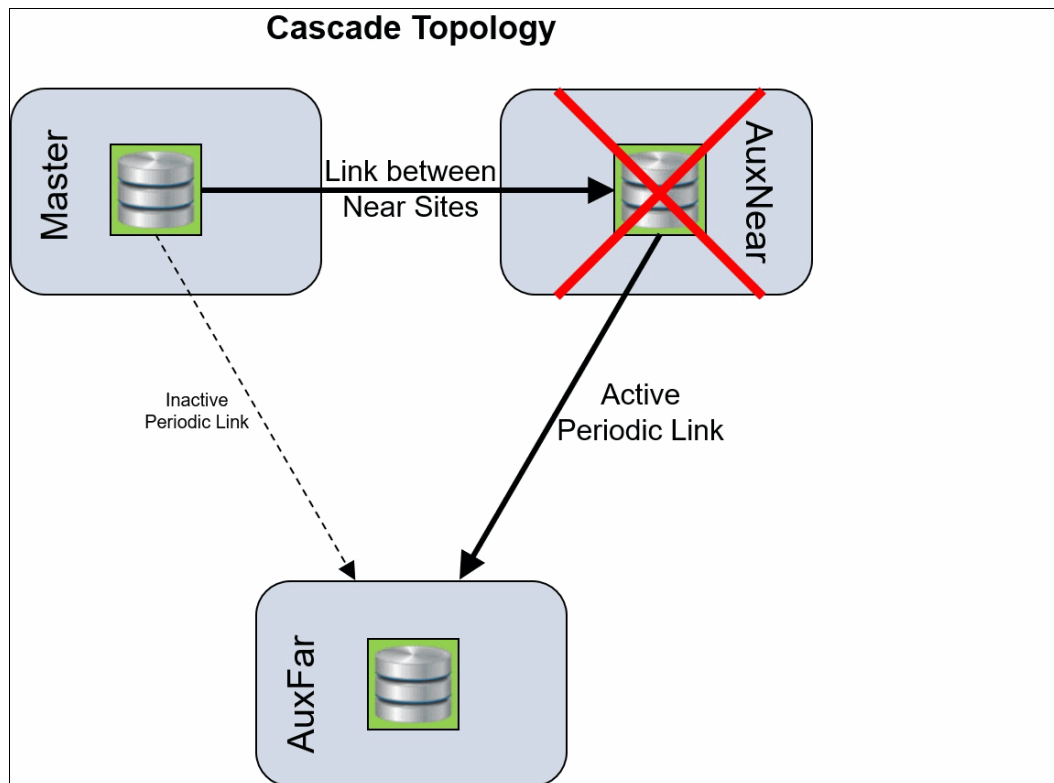


Figure 6-15 AuxNear site failure in a Cascade topology

1. Issue the **ls3sitercconsistgrp** command to verify the state of the Consistency Groups. In the resulting output, the state of the Consistency Group is stopped.

Example 6-17 Verify the current state

```
[root@localhost ~]# ls3sitercconsistgrp
name state primary periodic_copy_source sync_copy_state periodic_copy_state ... status
CGO stopped Master AuxNear idling_disconnected copied ... periodic_src_offline
```

2. Temporarily change the periodic source from the failed AuxNear site to the remaining AuxNear site. This option provides an up-to-date second (asynchronous) data copy although the AuxNear site is not available. You may also choose to leave 3-Site cycling inactive for a period while the AuxNear site experiences problems. For this, skip this step and also step #6.

To change the periodic source and start data cycling in the 2site_periodic state, use the **ch3sitercconsistgrp -periodicsrc** command as shown in Example 6-18.

Example 6-18 Change periodic source and start cycling

```
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc Master CGO
[root@localhost ~]# ls3sitercconsistgrp
```

name	state	primary	periodic_copy_source	sync_copy_state	periodic_copy_state	freeze_time	cycle_period	status
CG0	2site_periodic	Master	Master	idling_disconnected	idling	2020/04/02/19/02/18	300	partner_offline

3. Perform recovery activities on the AuxNear site to restore it to a fully functional state.
4. After AuxNear is recovered, check the `sync_copy_state` in the `ls3sitercconsistgrp` output:
 - If it is `consistent_synchronized`, the site is automatically included in data cycling.
 - If it is `consistent_stopped`, issue the `ch3sitercconsist -join` command to include the site in data cycling, as shown in Example 6-19.

Example 6-19 Include the site in data cycling

```
[root@localhost ~]# ch3sitercconsistgrp -join AuxNear CG0
```

5. Issue the `ls3sitercconsistgrp` command to verify the recovery, as shown in Example 6-20.

Example 6-20 Verify the recovery

```
[root@localhost ~]# ls3sitercconsistgrp
```

name	state	primary	periodic_copy_source	sync_copy_state	periodic_copy_state	freeze_time	cycle_period	status
CG0	3site_consistent	Master	Master	consistent_synchronized	stopped	2020/04/02/19/27/48	300	online

6. Change the setup back to a cascade topology by switching the periodic source, as shown in Example 6-21.

Example 6-21 Change the configuration back to Cascade topology and verify

```
[root@localhost ~]# ch3sitercconsistgrp -periodicsrc AuxNear CG0
```

```
[root@localhost ~]# ls3sitercconsistgrp
```

name	state	primary	periodic_copy_source	sync_copy_state	periodic_copy_state	freeze_time	cycle_period	status
CG0	3site_consistent	Master	AuxNear	consistent_synchronized	stopped	2020/04/02/19/41/08	300	online

6.3.5 Recovery from a disaster at the primary site (scenarios #3 and #5)

If the primary system fails and read/write access to application data is required, you must use the surviving AuxNear site GUI or CLI to enable access to the remaining Metro Mirror secondary site.

In both 3-Site topologies, recovery is similar. An example for a star topology is shown in Figure 6-16 and described below. For a cascade topology, the same steps need to be followed, with the exception of the periodic source switch.

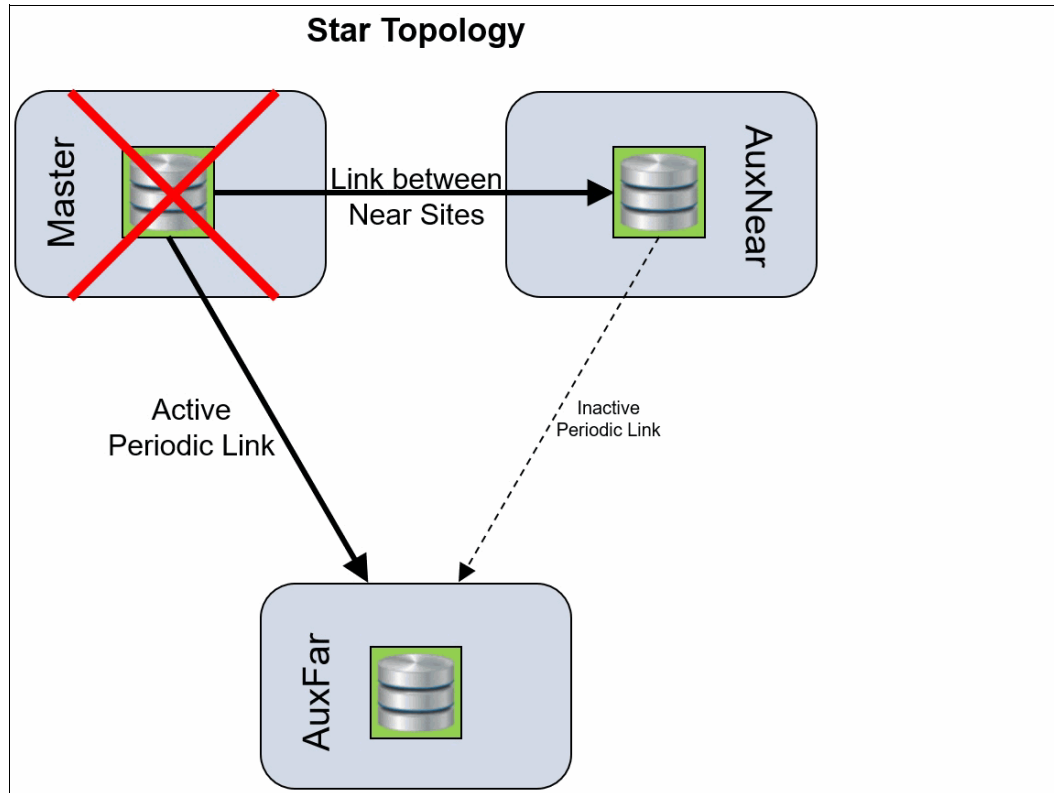


Figure 6-16 Master site failure in a Star topology

To recover from a disaster at the Master site, complete the following steps:

1. Issue the `ls3sitercconsistgrp` command to verify the state of the Consistency Groups. In the resulting output, the state of the Consistency Group is stopped. Example 6-22 shows the status `primary_and_periodic_src_offline` in a star topology.

Example 6-22 Verify Consistency Group status

```
[root@localhost ~]# ls3sitercconsistgrp
name state  primary periodic_copy_source sync_copy_state      periodic_copy_state ... status
CG0  stopped Master  Master                consistent_disconnected None                ... primary_and_periodic_src_offline
```

Example 6-23 shows the status `primary_offline` in a cascade topology.

Example 6-23 Verify Consistency Group status

```
[root@localhost ~]# ls3sitercconsistgrp
name state  primary periodic_copy_source sync_copy_state      periodic_copy_state ... status
CG0  stopped Master  AuxNear                consistent_disconnected stopped                ... primary_offline
```

2. If the primary site or storage fails, access to application data is disrupted. Until the primary is recovered, you have the following options:
 - Option A is to not run any application I/O on the surviving site and the Consistency Groups remain in the stopped state.
 - Option B is to enable read/write access to volume copies on the surviving AuxNear site by using Metro Mirror commands in the GUI or CLI of the surviving site. For detailed instructions, refer to *IBM System Storage SAN Volume Controller and Storwize V7000 Replication Family Services*, SG24-7574.

3. If Option A was chosen, then after the Master site is recovered the 3-Site configuration will either recover automatically, or it will require joining. Check the `sync_copy_state` in `ls3siterconsistgrp` output:
 - If it is `consistent_synchronized`, the site is automatically included in data cycling.
 - If it is `consistent_stopped`, issue the `ch3siterconsist -join` command to include the site in data cycling.
 4. If Option B was chosen, then after access to volumes on the surviving site is enabled, 3-Site Consistency Groups in a cascade topology will switch to `2site_periodic` state. Consistency Groups in a star topology will need the periodic source to be switched using the `ch3siterconsistgrp -periodicsrc` command to get to `2site_periodic` state.
- After the Master is fully recovered, join the site as it was previously and switch the primary and the periodic source back to the intended configuration.

6.3.6 Recovery from dual site failures

A site loss or a storage system failure can be a temporary or permanent issue and in this chapter we only show the concept of how to perform recovery operations.

To recover from dual disasters at both near sites as shown in Figure 6-17, complete the following high level steps.

Important: You must contact IBM Support before attempting this procedure.

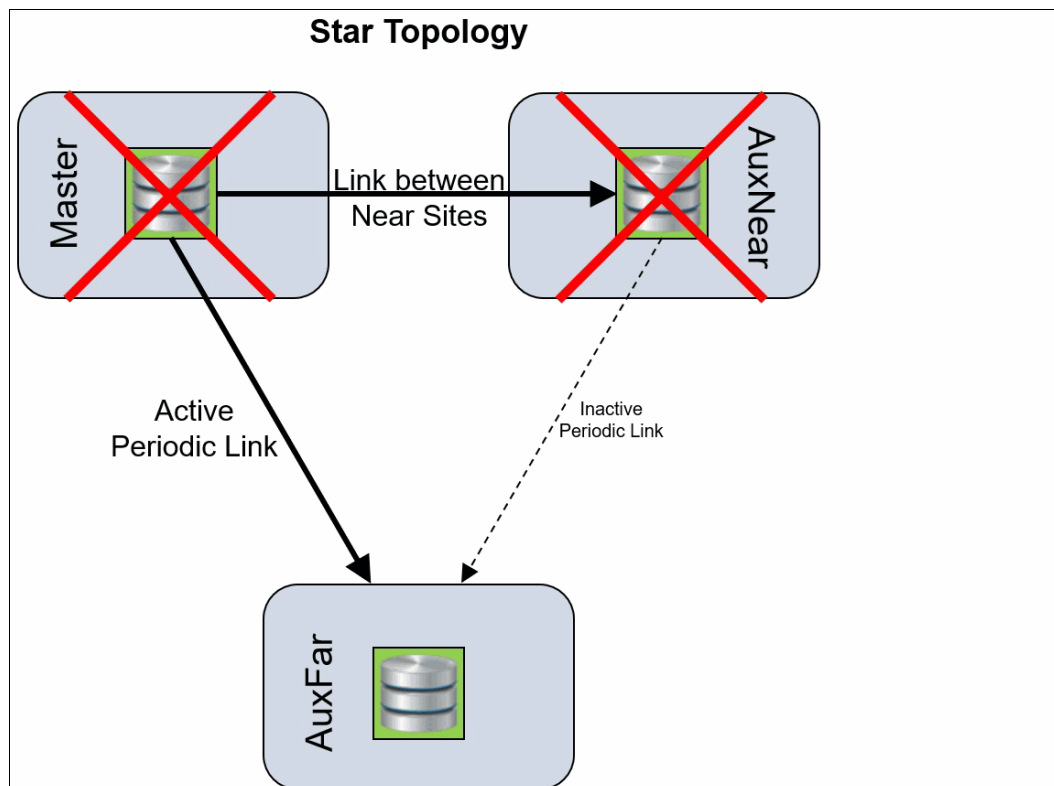


Figure 6-17 Two near site failures

1. Verify that when the condition occurs the 3-Site Consistency Groups are in the stopped state.

2. Issue the **converttrcconsistgrp** command to make all the AuxFar site volumes accessible and convert the 3-Site Consistency Group to a 2-Site Consistency Group.
3. Recover both near sites (Master and AuxNear) using the manual procedure.
4. Optionally, you can recover data from the AuxFar site default volume to a new volume on either of the two near sites using a 2-Site remote copy service.
5. Optionally, you can again move to 3-Site data cycling by following the standard procedure of converting volumes to a 3-Site Consistency Group. Once a synchronized copy of data is available on the AuxFar site new volumes, delete the old volumes.

Important: You *must* contact IBM Support before attempting this procedure, and certainly before you attempt to recover from a dual disaster at any other two sites.

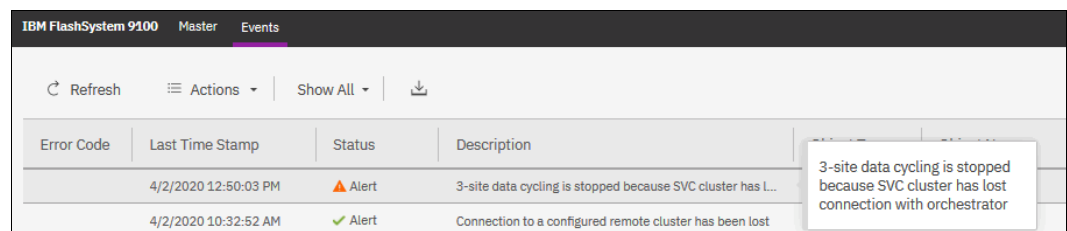
6.4 Recovering from 3-Site Orchestrator failure

3-Site Orchestrator is a key component of the 3-Site Replication solution. The steps required to recover from a failure of a link between 3-Site Orchestrator, and any of the storage systems, and to recover from a failure of a system that hosts 3-Site Orchestrator, are described in this section.

6.4.1 3-Site Orchestrator link failure

If the 3-Site Orchestrator link (SSH connection to any site) fails, complete the following steps:

1. Verify that the 3-Site data cycling is stopped because the system has lost connection with the 3-Site Orchestrator. Verify the event notification using the event log. The event is shown in the event logs of all three clusters. One example is shown in Figure 6-18. This event clears automatically when the secure shell (SSH) connection is restored.



Error Code	Last Time Stamp	Status	Description
	4/2/2020 12:50:03 PM	Alert	3-site data cycling is stopped because SVC cluster has lost connection with orchestrator
	4/2/2020 10:32:52 AM	Alert	Connection to a configured remote cluster has been lost

Figure 6-18 Lost Orchestrator Connection

2. If the link failure occurs before creating 3-Site Consistency Groups:
 - Recover the SSH connections to any site.
3. If the link failure occurs after 3-Site Consistency Groups are created:

Consider the following information:

- The Consistency Groups move to the stopped state.
- Data cycling automatically restarts when the SSH connections to any site are recovered.
- All the commands except the **converttrcconsistgrp** command return the error shown in Example 6-24:

Example 6-24 Error message during 3-Site Orchestrator disconnect

```
CMMVC9507E SSH Connection failed with site [Master_cluster_ip]
```

6.4.2 Recovery from a 3-Site Orchestrator down situation

To migrate or reinstall 3-Site Orchestrator complete the steps described in this procedure.

A 3-Site Orchestrator migration might be necessary if the host system fails and cannot be recovered, or if it becomes necessary to migrate the 3-Site Orchestrator instance to another host.

It might be necessary to reinstall 3-Site Orchestrator if the 3-Site Orchestrator installation is corrupted, the host fails, or a similar catastrophic event occurs.

1. Verify the installation port for the 3-Site Orchestrator.
2. Stop (or uninstall if required) the 3-Site Orchestrator package to shutdown the active 3-Site Orchestrator instance.
3. Install the 3-Site Orchestrator package on the new host where migration of the host is expected.
4. Configure secure shell (SSH) key-based passwordless authentication between the new host and the systems.
5. Create the 3-Site Orchestrator configuration using the **mk3siteconfig** command. All three storage systems must be available at this point in time.
6. After the configuration is complete, restart the 3-Site Orchestrator host. This runs recovery of all the 3-Site objects on the target 3-Site Orchestrator host and starts data cycling for all available 3-Site Orchestrator Consistency Groups.

A new 3-Site Orchestrator can only be installed if all three sites are reachable using SSH. This might not be the case in a DR situation, therefore a standby 3-Site Orchestrator should be planned and installed.

Related publications

The publications listed in this section are considered particularly suitable for a more detailed discussion of the topics covered in this book.

IBM Redbooks

The following IBM Redbooks publications provide additional information about the topic in this document. Note that some publications referenced in this list might be available in softcopy only.

- ▶ *Implementing the IBM SAN Volume Controller with IBM Spectrum Virtualize V8.3.1*, SG24-8465
- ▶ *Implementing the IBM FlashSystem 9200, 9100, 7200 and 5100 with IBM Spectrum Virtualize V8.3.1*, SG24-8466

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Glossary

3-Site Orchestrator Software application, that runs on a Linux host and is responsible for coordination and management of 3-Site replications.

Access point Vdisk object, designated to a volume in 3-Site replication relationship and linked to it with forward and reverse FlashCopy mappings.

Consistency Group A container for remote-copy relationships. When relationships are in a Consistency Group, they can't change states independently of each other. Individual mappings in a Consistency Group can't be started or stopped separately. Management operations are performed on the entire group instead of the individual mappings.

FlashCopy A function that creates a point-in-time copy of data that is stored on a source volume to a target volume.

Link

Active Periodic Link (or Active Link) A link to the auxiliary-far site from the master site or auxiliary-near site that is responsible for asynchronous replication of the data between the sites (periodic replication).

Metro Mirror Link A link between master site and auxiliary-near site that is used for synchronous replication.

Inactive Periodic Link (or Inactive Link, Standby Link) A link to the auxiliary-far site from the master site or auxiliary-near site. The link is in an idle state. The link is available to be activated in case of an active link failure.

Metro Mirror A type of remote copy that creates a synchronous copy of data from a primary volume to a secondary volume.

Partnership An association between two Spectrum Virtualize systems. Systems in a partnership can use Remote Copy functions to replicate data between them.

Periodic source A site which holds the source volumes for periodic replication to AuxFar site. Location of a periodic source is determined by the 3-Site topology: with the star topology, the primary and periodic source sites match, with the cascaded topology, the primary site replicates data to AuxNear, which serves as a periodic source.

Relationship (remote copy relationship, RC relationship) An association between volumes (vdisk) on partnered systems. Data is replicated between volumes in a remote copy relationship.

Site

Auxiliary-far site (or AuxFar) A data retention site that stores a third copy of mirrored data. It is a target for periodic replication and is typically 100 km or more from the near sites.

Auxiliary-near site (or AuxNear, Near Disaster Recovery site) A site that contains a mirror copy of the data on the master site. The distance between the master site and this site must be within a range to be able to establish Metro Mirror remote copy services.

Master site (or Production site, Primary site) A site where applications typically are run.

Near sites A name for Master and AuxNear sites together.

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